

Performance Analysis

 Reattempt

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Detailed analysis of your performance in the AWS Certified Solutions Architect Associate - SAA C03 exam.

 Answer Distribution

0
Correct

0
Incorrect

65
Unattempted

 Domain Performance

Cost Optimization	0%
Databases	0%
Storage	0%
Compute	0%
Networking	0%
Security	0%
Monitoring and Observability	0%
Application Integration	0%



Passing score: 48

Time Taken



130m 0s

Total time: 130 minutes

Question Analysis

✓ Correct

0

✗ Incorrect

0

⊖ Unattempted

65

🔖 Reviewed

0

Question-wise Analysis

⌕ Expand All

⌕ Collapse All

Question 1

Not Attempted

Cost Optimization



A company stores 300 TB of data in Amazon S3 Standard storage. The S3 objects, around 50 GB each, are frequently replaced using multipart uploads by a global application. Although the data volume and object size remain consistent, S3 storage costs are increasing. What strategy should a solutions architect employ to reduce these costs?

- ⊗ Switch from multipart uploads to Amazon S3 Transfer Acceleration.
Switching to S3 Transfer Acceleration will not directly address increasing costs related to incomplete multipart uploads. It focuses on accelerating data transfers, not reducing storage overhead.
- ✓ Enable an S3 Lifecycle policy that deletes incomplete multipart uploads.
This is the correct approach. Incomplete multipart uploads consume storage space without providing accessible objects. Configuring an S3 Lifecycle policy to automatically delete these incomplete uploads will reduce storage costs.
- ⊗ Configure S3 inventory to prevent objects from being archived too quickly.
S3 Inventory is used to manage and audit the replication and encryption status of your objects for business, compliance and regulatory needs. Configuring S3 inventory does not directly address costs associated with multipart uploads, archiving is not a concern in this scenario.
- ⊗ Configure Amazon CloudFront to reduce the number of objects stored in Amazon S3.
Amazon CloudFront is a CDN and won't directly influence the number of objects stored in S3, so this will not help the cost increase caused by uncompleted multipart uploads.

Overall Explanation:

The increasing costs are most likely due to incomplete multipart uploads consuming storage without being finalized into complete objects. Enabling an S3 Lifecycle policy to automatically delete incomplete multipart uploads is the most effective solution to reduce these costs. S3 Transfer acceleration will improve upload speed but does not reduce storage cost. S3 Inventory and CloudFront will not help reduce cost associated with multipart uploads.

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Question 2

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[Databases](#)

A company has deployed a multiplayer game for mobile devices that requires live location tracking of players based on latitude and longitude. The game uses an Amazon RDS for PostgreSQL DB instance with read replicas to store the location data. During peak usage periods, the database struggles to maintain acceptable performance for reading and writing location updates, and the game's user base is growing rapidly. What should a solutions architect do to improve the performance of the data tier in this scenario?

- ⊗ Take a snapshot of the existing DB instance. Restore the snapshot with Multi-AZ enabled.

Enabling Multi-AZ improves availability but doesn't directly address performance bottlenecks related to rapid updates and retrieval.

- ⊗ Migrate from Amazon RDS to Amazon OpenSearch Service with OpenSearch Dashboards.

While OpenSearch is good for search and analytics, migrating the database might not be the best solution for rapid, transactional updates of location data.

- ⊗ Deploy Amazon DynamoDB Accelerator (DAX) in front of the existing DB instance. Modify the game to use DAX.

DAX is specifically designed for DynamoDB. It does not work with Amazon RDS PostgreSQL instances. Thus, it is not a relevant solution in this case.

- ✓ Deploy an Amazon ElastiCache for Redis cluster in front of the existing DB instance. Modify the game to use Redis.

ElastiCache for Redis provides an in-memory data store with very low latency, suitable for caching frequently accessed data and handling high-volume read and write operations. Using Redis as a cache for location data can significantly improve the game's performance.

Overall Explanation:

ElastiCache for Redis offers a low-latency, in-memory caching solution that can improve read and write performance for location data updates. It is well-suited for high-volume, real-time applications like multiplayer games. While the existing database is PostgreSQL, ElastiCache can be added in front to serve as a fast cache for location data, which is more effective than other options like Multi-AZ or migrating the database entirely.

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Question 3

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Databases



A company stores critical data in Amazon DynamoDB tables in the company's AWS account. An IT administrator accidentally deleted a DynamoDB table, causing a significant loss of data and disruption to the company's operations. The company wants to prevent this type of disruption in the future with the LEAST operational overhead. Which solution will meet this requirement?

- ⊗ Configure a trail in AWS CloudTrail. Create an Amazon EventBridge rule for delete actions. Create an AWS Lambda function to automatically restore deleted DynamoDB tables.

This solution involves setting up several services (CloudTrail, EventBridge, Lambda), increasing operational overhead. It also assumes the automatic restoration is feasible and complete, which might not always be the case.

- ⊗ Create a backup and restore plan for the DynamoDB tables. Recover the DynamoDB tables manually.

While a good practice, this doesn't prevent accidental deletion. It only helps in recovery. Manual recovery also increases operational overhead during an incident.

- ✔ Configure deletion protection on the DynamoDB tables.

Deletion protection directly prevents accidental deletion of DynamoDB tables. It's a simple, built-in feature that adds minimal operational overhead.

- ⊗ Enable point-in-time recovery on the DynamoDB tables.

Point-in-time recovery allows you to restore a table to a previous state but doesn't prevent accidental deletion. It's a recovery mechanism, not a preventative measure.

Overall Explanation:

Configuring deletion protection on the DynamoDB tables directly prevents accidental deletions with the least operational overhead. It is a built-in feature designed precisely for this purpose. Other options involve more complex setups or only focus on recovery after a deletion has already occurred. <https://aws.amazon.com/about-aws/whats-new/2023/03/amazon-dynamodb-table-deletion-protection/>

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Question 4

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Storage



A company has an on-premises data center that is running out of storage capacity. The company wants to migrate its storage infrastructure to AWS while minimizing bandwidth costs. The solution must allow for immediate retrieval of data at no additional cost. How can these requirements be met?

- ⊗ Deploy Amazon S3 Glacier Vault and enable expedited retrieval. Enable provisioned retrieval capacity for the workload.

S3 Glacier is designed for archival data and involves costs for expedited retrieval. It's not suitable for immediate retrieval at no additional cost.
- ✔ Deploy AWS Storage Gateway using cached volumes. Use Storage Gateway to store data in Amazon S3 while retaining copies of frequently accessed data subsets locally.

AWS Storage Gateway in cached mode stores the primary data in S3, while frequently accessed data is cached on-premises for low-latency access. This minimizes the on-premises storage footprint and reduces bandwidth costs by only transferring frequently accessed data. It also allows for immediate retrieval of frequently accessed data.
- ⊗ Deploy AWS Storage Gateway using stored volumes to store data locally. Use Storage Gateway to asynchronously back up point-in-time snapshots of the data to Amazon S3.

While stored volumes offer low-latency access by storing all data locally, it doesn't address the on-premises storage capacity issue. Also, since they want to migrate to AWS, storing all data locally doesn't accomplish the migration goal.
- ⊗ Deploy AWS Direct Connect to connect with the on-premises data center. Configure AWS Storage Gateway to store data locally. Use Storage Gateway to asynchronously back up point-in-time snapshots of the data to Amazon S3.

AWS Direct Connect addresses network connectivity but doesn't directly address the storage capacity issue or minimize bandwidth costs related to immediate data retrieval. Storing all data locally defeats the purpose of migrating to AWS.

Overall Explanation:

AWS Storage Gateway using cached volumes is the most suitable solution. It minimizes on-premises storage, reduces bandwidth costs by caching only frequently accessed data locally, and enables immediate retrieval of frequently accessed data. Data is primarily stored in Amazon S3, aligning with the migration goal. Stored volumes do not address storage capacity issue and Glacier expedited retrieval comes with additional cost.

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Question 5

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[Compute](#)

A company hosts a three-tier web application within a VPC across multiple Availability Zones. The application tier consists of Amazon EC2 instances managed by an Auto Scaling group. The

company requires an automated scaling plan that analyzes the daily and weekly historical workload trends of each resource, ensuring appropriate scaling based on both forecasted demand and real-time utilization changes. Which scaling strategy should a solutions architect recommend to fulfill these requirements?

- ☐ Implement dynamic scaling with step scaling based on average CPU utilization from the EC2 instances.

This option only addresses dynamic scaling based on a single metric (CPU utilization) and does not incorporate historical trend analysis for forecasting.

- ☒ Enable predictive scaling to forecast and scale. Configure dynamic scaling with target tracking.

This option combines predictive scaling, which analyzes historical trends, with dynamic scaling and target tracking, which responds to real-time utilization changes. This is the most appropriate choice.

- ☐ Create an automated scheduled scaling action based on the traffic patterns of the web application.

Scheduled scaling is based on known, repeating patterns, but it does not dynamically adjust to unexpected changes in utilization or incorporate historical trend analysis.

- ☐ Set up a simple scaling policy. Increase the cooldown period based on the EC2 instance startup time.

This option focuses on optimizing the cooldown period, not on implementing a scaling strategy based on both forecasting and real-time utilization.

Overall Explanation:

The most suitable solution is to enable predictive scaling, which forecasts future demand based on historical trends, and combine it with dynamic scaling using target tracking, which responds to real-time changes in resource utilization. This approach ensures that the application scales appropriately based on both predicted demand and current load.

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Question 6

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Networking



A company recently launched Linux-based application instances on Amazon EC2 in a private subnet and launched a Linux-based bastion host on an Amazon EC2 instance in a public subnet

of a VPC. A solutions architect needs to connect from the on-premises network, through the company's internet connection, to the bastion host, and to the application servers. The solutions architect must make sure that the security groups of all the EC2 instances will allow that access. Which combination of steps should the solutions architect take to meet these requirements? (Choose two.)

- ☐ Replace the current security group of the bastion host with one that only allows inbound access from the application instances.

This option is incorrect because the bastion host needs to allow inbound connections from the on-premises network, not the application instances.

- ☐ Replace the current security group of the bastion host with one that only allows inbound access from the internal IP range for the company.

This option is incorrect because the connection from on-premise to the Bastion host goes through the internet and will not be using the internal IP range.

- ☒ Replace the current security group of the bastion host with one that only allows inbound access from the external IP range for the company.

This option is correct. This will restrict access to the bastion host from the specific IP range of the on-premises network, ensuring secure connectivity. It ensures that only authorized users from the on-premises network can access the bastion host.

- ☒ Replace the current security group of the application instances with one that allows inbound SSH access from only the private IP address of the bastion host.

This option is correct. This step enables SSH connectivity from the bastion host to the application instances in the private subnet. By allowing inbound SSH access only from the private IP address of the bastion host, you ensure that SSH access is restricted to the bastion host only.

- ☐ Replace the current security group of the application instances with one that allows inbound SSH access from only the public IP address of the bastion host.

This option is incorrect. Since the application instances are in a private subnet, allowing inbound SSH from the bastion host's public IP is not the ideal approach.

Overall Explanation:

The correct options are C and D. To connect from on-premises to the application instances via the bastion host, the bastion host's security group must allow inbound access from the company's external IP range (option C). The application instances' security group must

allow inbound SSH access only from the bastion host's private IP address (option D), ensuring secure access within the VPC.

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Question 7

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Databases



A package delivery company's application uses Amazon EC2 instances and an Amazon Aurora MySQL DB cluster. As the application gains popularity, EC2 instance usage increases slightly, but DB cluster usage increases at a much faster rate. The company added a read replica, which temporarily reduced DB cluster usage, but the load continues to grow. Repeated read statements related to delivery details are the primary cause of the increased DB cluster usage. The company needs to reduce the impact of these repeated reads on the DB cluster MOST cost-effectively. Which solution should the company implement?

- ☒ Implement an Amazon ElastiCache for Redis cluster between the application and the DB cluster.

ElastiCache for Redis can cache the results of repeated read queries, reducing the load on the Aurora DB cluster and improving performance. Caching is an efficient way to alleviate the pressure from repeated read operations, making it a cost-effective solution.

- ☐ Add an additional read replica to the DB cluster.

While adding read replicas can distribute the read load, it does not address the underlying issue of repeated identical queries. It may temporarily reduce load but won't prevent the database from processing the same requests repeatedly, making it less cost-effective than caching.

- ☐ Configure Aurora Auto Scaling for the Aurora read replicas.

Aurora Auto Scaling dynamically adjusts the number of read replicas based on load. While this can help with scaling, it doesn't prevent repeated identical read requests from reaching the database in the first place. Therefore, it is not as cost-effective as implementing a caching layer.

- ☐ Modify the DB cluster to have multiple writer instances.

Aurora MySQL does not support multiple writer instances. Also, the problem described is read-heavy. Thus, adding writer instances would not solve the problem.

Overall Explanation:

The most cost-effective solution is to implement an Amazon ElastiCache for Redis cluster. This approach caches the results of repeated read statements, reducing the load on the Aurora DB cluster. While read replicas can help, they don't prevent the database from processing the same repeated requests. Aurora Auto Scaling helps with scaling, but it doesn't address the repeated nature of the read requests like a caching solution does. Aurora MySQL does not support multiple writer instances.

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Question 8

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Databases



A company's application uses an Amazon DynamoDB table. Many requests aren't returning the latest data, but other database performance metrics are normal. What design change should a solutions architect recommend?

- ☐ Add read replicas to the table.
Adding read replicas won't solve data staleness issues. Read replicas typically improve read throughput but don't guarantee data consistency.
- ☐ Use a global secondary index (GSI).
Using a GSI won't fix the inconsistency problem. GSIs are used to optimize query performance with different access patterns.
- ☒ Request strongly consistent reads for the table.
DynamoDB defaults to eventual consistency, which means reads might not reflect the latest writes. Requesting strongly consistent reads ensures that the latest data is returned.
- ☐ Request eventually consistent reads for the table.
The application is already using eventually consistent reads, which is the root cause of the problem.

Overall Explanation:

DynamoDB defaults to eventual consistency for read operations. To ensure that read operations return the latest data and address the issue of stale data being returned to users, the solutions architect should recommend switching the read consistency level from eventually consistent reads to strongly consistent reads.

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Question 9

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Security



A company has deployed its application on Amazon EC2 instances with an Amazon RDS database. The company used the principle of least privilege to configure the database access credentials. The company's security team wants to protect the application and the database from SQL injection and other web-based attacks with the LEAST operational overhead.

☐ Use security groups and network ACLs to secure the database and application servers.
Security groups and network ACLs are helpful for network-level access control, but they don't directly protect against SQL injection and other web-based attacks.

☒ Use AWS WAF to protect the application. Use RDS parameter groups to configure the security settings.

AWS WAF is a web application firewall that helps protect your web applications from common web exploits and bots. Parameter groups can help configure security settings for the database such as auditing.

☐ Use AWS Network Firewall to protect the application and the database.

AWS Network Firewall is a network-level firewall service. While it offers protection, it is less specialized for web application attacks like SQL injection compared to WAF and might involve higher operational overhead for application-level protection.

☐ Use different database accounts in the application code for different functions. Avoid granting excessive privileges to the database users.

While using different database accounts with least privilege is a good security practice, it doesn't prevent SQL injection attacks. It limits the impact if an attack is successful, but doesn't stop the attack itself.

Overall Explanation:

AWS WAF (Web Application Firewall) is specifically designed to protect web applications from attacks like SQL injection. Using AWS WAF provides the best protection against the specified attacks with the least operational overhead. RDS parameter groups can be used to configure database security settings. Network ACLs and security groups provide network-level security, while database user separation with least privilege is a general

security best practice but neither directly address SQL injection attacks as effectively as WAF. Therefore, the correct answer is B.

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Question 10

Not Attempted

Security



An ecommerce company runs applications in AWS accounts that are part of an organization in AWS Organizations. The applications run on Amazon Aurora PostgreSQL databases across all the accounts. The company needs to prevent malicious activity and must identify abnormal failed and incomplete login attempts to the databases. Which solution will meet these requirements in the MOST operationally efficient way?

- ☐ Attach service control policies (SCPs) to the root of the organization to identify the failed login attempts.
Attaching SCPs won't directly identify failed login attempts. SCPs control permissions and don't analyze login activity.
- ☒ Enable the Amazon RDS Protection feature in Amazon GuardDuty for the member accounts of the organization.
GuardDuty's RDS Protection feature directly monitors RDS login activity, detects suspicious behavior, and is designed for this specific use case. It is operationally efficient because it's a managed service.
- ☐ Publish the Aurora general logs to a log group in Amazon CloudWatch Logs. Export the log data to a central Amazon S3 bucket.
While this provides the logs, it requires significant manual effort to analyze the data and set up alerts, making it less operationally efficient.
- ☐ Publish all the Aurora PostgreSQL database events in AWS CloudTrail to a central Amazon S3 bucket.
CloudTrail doesn't capture all the details of database login attempts. Additionally, analyzing the large volume of CloudTrail data for specific login attempts would be operationally complex.

Overall Explanation:

Amazon GuardDuty's RDS Protection feature continuously monitors RDS login activity and profiles it for suspicious activities like unauthorized access or unusual login patterns. It's

the most operationally efficient solution because it's a managed service specifically designed for threat detection in RDS environments. Options involving CloudWatch Logs or CloudTrail require extensive manual configuration and analysis, making them less efficient. SCPs are for permission control, not threat detection.

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Question 11

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Networking



A company has an AWS Direct Connect connection from its corporate data center to its VPC in the us-east-1 Region. The company acquired another corporation with several VPCs and a Direct Connect connection between its on-premises data center and the eu-west-2 Region. The CIDR blocks for the VPCs of the company and the corporation do not overlap. The company requires connectivity between the two Regions and the data centers with a solution that is scalable and reduces operational overhead. What should a solutions architect do to meet these requirements?

- ⊗ Set up inter-Region VPC peering between the VPC in us-east-1 and the VPCs in eu-west-2.

Inter-Region VPC peering does not natively extend to on-premises data centers through Direct Connect. This solution would not provide connectivity for on-premises resources.

- ⊗ Create private virtual interfaces from the Direct Connect connection in us-east-1 to the VPCs in eu-west-2.

Direct Connect virtual interfaces are region-specific. You cannot directly connect a virtual interface from one region's Direct Connect to VPCs in another region.

- ⊗ Establish VPN appliances in a fully meshed VPN network hosted by Amazon EC2. Use AWS VPN CloudHub to send and receive data between the data centers and each VPC. While this option could work, it introduces operational overhead by requiring the management of EC2 instances and VPN appliances. It may also not be as scalable or cost-effective as other solutions.

- ✔ Connect the existing Direct Connect connections to a Direct Connect gateway. Route traffic from the virtual private gateways of the VPCs in each Region to the Direct Connect gateway.

A Direct Connect gateway provides a global point for Direct Connect traffic. It allows connecting Direct Connect connections from different regions to VPCs in different regions through virtual private gateways (VGWs). This solution is scalable and reduces operational overhead.

Overall Explanation:

The best solution is to connect the existing Direct Connect connections to a Direct Connect gateway. By routing traffic from the virtual private gateways of the VPCs in each Region to the Direct Connect gateway, the company can achieve scalable inter-region connectivity with reduced operational overhead. A Direct Connect gateway acts as a central hub for routing traffic between VPCs in different regions through the Direct Connect connections, and it's designed for scenarios involving multiple regions and Direct Connect connections.

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Question 12

Not Attempted

Databases



A mobile gaming company requires a solution to handle high traffic spikes and process score updates in order of receipt for a leaderboard. The processed updates must be stored in a highly available database with minimal operational overhead. Which architecture best meets these requirements?

- ✔ Push score updates to Amazon Kinesis Data Streams. Process the updates in Kinesis Data Streams with AWS Lambda. Store the processed updates in Amazon DynamoDB.
- This option satisfies all requirements: Kinesis Data Streams handles traffic spikes and ordered processing, Lambda provides serverless processing, and DynamoDB is a highly available, managed NoSQL database with minimal overhead.
- ✘ Push score updates to Amazon Kinesis Data Streams. Process the updates with a fleet of Amazon EC2 instances set up for Auto Scaling. Store the processed updates in Amazon Redshift.
- While Kinesis Data Streams handles traffic spikes and ordered processing, managing EC2 instances introduces operational overhead. Amazon Redshift is not designed for single record updates, which are required in this use case.

- ⊗ Push score updates to an Amazon Simple Notification Service (Amazon SNS) topic. Subscribe an AWS Lambda function to the SNS topic to process the updates. Store the processed updates in a SQL database running on Amazon EC2.
SNS does not guarantee message ordering. Using a SQL database on EC2 increases operational overhead compared to DynamoDB.
- ⊗ Push score updates to an Amazon Simple Queue Service (Amazon SQS) queue. Use a fleet of Amazon EC2 instances with Auto Scaling to process the updates in the SQS queue. Store the processed updates in an Amazon RDS Multi-AZ DB instance.
While SQS can handle traffic spikes, it does not inherently guarantee strict ordering. Using EC2 instances and RDS Multi-AZ for processing and storage introduces more operational overhead compared to using Lambda and DynamoDB.

Overall Explanation:

The optimal solution uses Amazon Kinesis Data Streams to ingest and maintain the order of score updates, AWS Lambda for serverless processing, and Amazon DynamoDB for highly available data storage with minimal operational overhead. Other options either lack ordering guarantees, introduce unnecessary operational overhead with EC2 instances and managed relational databases, or are not suited for single record updates.

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Question 13

Not Attempted

[Monitoring and Observability](#)

A company has multiple AWS accounts with applications deployed in the us-west-2 Region. Application logs are stored within Amazon S3 buckets in each account. The company wants to build a centralized log analysis solution that uses a single S3 bucket. Logs must not leave us-west-2, and the company wants to incur minimal operational overhead. Which solution meets these requirements and is MOST cost-effective?

- ⊗ Create an S3 Lifecycle policy that copies the objects from one of the application S3 buckets to the centralized S3 bucket.
S3 Lifecycle policies are primarily used for transitioning objects between storage classes, not for copying objects between buckets. This is not the most suitable approach for log aggregation.
- ✓ Use S3 Same-Region Replication to replicate logs from the S3 buckets to another S3 bucket in us-west-2. Use this S3 bucket for log analysis.

S3 Same-Region Replication (SRR) is a cost-effective solution that automatically and asynchronously replicates objects between S3 buckets in the same AWS Region. It minimizes operational overhead and keeps the data within the us-west-2 Region.

- ✘ Write a script that uses the PutObject API operation every day to copy the entire contents of the buckets to another S3 bucket in us-west-2. Use this S3 bucket for log analysis.

Writing a script to copy logs daily using the PutObject API introduces significant operational overhead for maintenance, error handling, and scheduling. It's less cost-effective than replication due to compute costs and manual effort.

- ✘ Write AWS Lambda functions in these accounts that are triggered every time logs are delivered to the S3 buckets (s3:ObjectCreated:* event). Copy the logs to another S3 bucket in us-west-2. Use this S3 bucket for log analysis.

Using Lambda functions triggered by S3 events introduces operational overhead and potential complexity in managing the functions across multiple accounts. While this would work, SRR is a simpler and more cost-effective choice.

Overall Explanation:

S3 Same-Region Replication (SRR) provides an automated, cost-effective, and operationally efficient way to centralize logs from multiple S3 buckets into a single bucket within the same region (us-west-2). SRR minimizes operational overhead compared to scripting or Lambda functions. Lifecycle policies are not designed for cross-bucket data transfer.

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Question 14

Not Attempted

Storage



A company delivers on-demand training videos to students globally, allowing authorized developers to upload content. Data is stored in an Amazon S3 bucket in us-east-2. The company has created S3 buckets in eu-west-2 and ap-southeast-1 and aims to replicate data to minimize latency for uploads near eu-west-2 and streaming near ap-southeast-1. Which combination of steps meets these requirements with the FEWEST application changes?

- ✘ Configure one-way replication from the us-east-2 S3 bucket to the eu-west-2 S3 bucket. Configure one-way replication from the us-east-2 S3 bucket to the ap-southeast-1 S3 bucket.

This setup replicates data from the primary region (us-east-2) to the new regions, but doesn't necessarily optimize for uploads from the new regions or keep the buckets synchronized if uploads occur in multiple regions.

- ⊗ Configure one-way replication from the us-east-2 S3 bucket to the eu-west-2 S3 bucket. Configure one-way replication from the eu-west-2 S3 bucket to the ap-southeast-1 S3 bucket.

This creates a chain replication which may introduce latency issues as replication has to go through eu-west-2 before reaching ap-southeast-1. It doesn't efficiently handle uploads from ap-southeast-1.

- ✓ Configure two-way (bidirectional) replication among the S3 buckets that are in all three Regions.

Two-way replication ensures that any changes made in any of the three S3 buckets are reflected in the other two. This addresses the scenario where developers might be uploading videos from locations closer to the eu-west-2 or ap-southeast-1 regions.

- ⊗ Create an S3 Multi-Region Access Point. Modify the application to use the Amazon Resource Name (ARN) of the Multi-Region Access Point for video streaming. Do not modify the application for video uploads.

This option improves streaming latency by directing video requests to the nearest S3 bucket via the Multi-Region Access Point, but does not address optimizing upload latency.

- ✓ Create an S3 Multi-Region Access Point. Modify the application to use the Amazon Resource Name (ARN) of the Multi-Region Access Point for video streaming and uploads.

Using a Multi-Region Access Point for both streaming and uploads allows the application to direct requests to the closest S3 bucket, minimizing latency for both operations.

Overall Explanation:

The best solution involves using an S3 Multi-Region Access Point for both streaming and uploads (Option E) and configuring two-way replication across all three S3 buckets (Option C). The Multi-Region Access Point ensures that users are directed to the nearest S3 bucket for both uploads and downloads, reducing latency. Two-way replication keeps all three S3 buckets synchronized, irrespective of where the initial upload occurs. This minimizes latency for both uploads and streaming with the fewest code changes to the application.

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Question 15

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Networking



A company has developed a new mobile application that allows users worldwide to view local news based on their selected topics and upload photos and videos directly through the app. Content is frequently accessed in the initial minutes after posting, with newer content quickly superseding older content, leading to the eventual disappearance of the older content. Given the local nature of the news, 90% of the content consumption occurs within the AWS Region where it was uploaded. Which solution will optimize the user experience by providing the LOWEST latency for content uploads?

- ☐ Upload and store content in Amazon S3. Use Amazon CloudFront for the uploads. CloudFront is primarily designed for content delivery (reading) and caching, not for optimizing upload latency.
- ☒ Upload and store content in Amazon S3. Use S3 Transfer Acceleration for the uploads. Amazon S3 Transfer Acceleration leverages CloudFront's globally distributed edge locations to accelerate content uploads to Amazon S3, providing the lowest latency.
- ☐ Upload content to Amazon EC2 instances in the Region that is closest to the user. Copy the data to Amazon S3. While this approach reduces latency for the initial upload, managing EC2 instances and then copying to S3 adds complexity and potential points of failure compared to S3 Transfer Acceleration.
- ☐ Upload and store content in Amazon S3 in the Region that is closest to the user. Use multiple distributions of Amazon CloudFront. While storing in the nearest S3 region is good, multiple CloudFront distributions are best for content delivery and do not directly optimize upload speed as effectively as S3 Transfer Acceleration.

Overall Explanation:

Amazon S3 Transfer Acceleration offers the lowest latency for content uploads by utilizing Amazon CloudFront's edge locations. Users upload to the nearest CloudFront edge, and the data is then transferred to the S3 bucket in the designated region using optimized network paths. This is more efficient than standard S3 uploads or using CloudFront distributions alone for upload optimization. Therefore, using S3 Transfer Acceleration will result in the best user experience in terms of upload speed.

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Question 16

Not Attempted

Application Integration



A company is developing a serverless application using an Amazon API Gateway REST API and AWS Lambda functions. The application needs to distribute messages received from the API Gateway to multiple target Lambda functions based on message content. The solution must filter messages to ensure each Lambda function receives only relevant data and minimize operational overhead. Which solution best meets these requirements?

- ☐ Send the requests from the API Gateway REST API to an Amazon Simple Notification Service (Amazon SNS) topic. Subscribe Amazon Simple Queue Service (Amazon SQS) queues to the SNS topic. Configure the target Lambda functions to poll the different SQS queues.

This option involves multiple services (SNS, SQS, and Lambda), increasing operational overhead. Lambda functions need to poll SQS queues, adding complexity.

- ☒ Send the requests from the API Gateway REST API to Amazon EventBridge. Configure EventBridge to invoke the target Lambda functions.

EventBridge can directly invoke Lambda functions based on filtering rules, offering a simpler and more efficient solution with less operational overhead. EventBridge rules can filter messages based on content, attributes, or patterns.

- ☐ Send the requests from the API Gateway REST API to Amazon Managed Streaming for Apache Kafka (Amazon MSK). Configure Amazon MSK to publish the messages to the target Lambda functions.

Amazon MSK is a complex streaming service that is not ideal for this use case. It introduces significant operational overhead compared to simpler solutions.

- ☐ Send the requests from the API Gateway REST API to multiple Amazon Simple Queue Service (Amazon SQS) queues. Configure the target Lambda functions to poll the different SQS queues.

While this provides message distribution, it lacks built-in filtering capabilities. Each Lambda would still need to filter, increasing function complexity, and adds overhead in managing multiple SQS queues.

Overall Explanation:

EventBridge provides a direct and efficient way to route messages to Lambda functions based on filtering rules, minimizing operational overhead compared to using SNS with SQS or directly using SQS queues. MSK introduces unnecessary complexity. While EventBridge has a default limit of 5 targets per rule, multiple rules can be created to bypass this limit.

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Question 17

Not Attempted

Security



A solutions architect is designing a two-tier web application. The application consists of a public-facing web tier hosted on Amazon EC2 instances in public subnets. The database tier consists of Microsoft SQL Server running on Amazon EC2 in a private subnet. Security is a high priority for the company. How should security groups be configured to ensure secure communication while adhering to the principle of least privilege? (Choose two.)

- ☒ Configure the security group for the web tier to allow inbound traffic on port 443 from 0.0.0.0/0.

This allows HTTPS traffic from any IPv4 address, enabling users to access the web application. While generally recommended to restrict the source IP range to a more specific range rather than allowing access from 0.0.0.0/0 (anywhere), it fulfills the requirement of public accessibility.

- ☐ Configure the security group for the web tier to allow outbound traffic on port 443 from 0.0.0.0/0.

While outbound traffic is allowed by default for security groups, explicitly opening outbound traffic to all destinations is not the most secure approach. Security groups are stateful, so return traffic for established connections is automatically allowed.

- ☒ Configure the security group for the database tier to allow inbound traffic on port 1433 from the security group for the web tier.

This allows the web tier EC2 instances to connect to the database tier on port 1433 (the default SQL Server port). By specifying the web tier's security group as the source, you restrict database access only to the web tier instances, enhancing security.

- ☐ Configure the security group for the database tier to allow outbound traffic on ports 443 and 1433 to the security group for the web tier.

Opening outbound ports on the database tier to the web tier on ports 443 and 1433 is unnecessary and a security risk. The database only needs to respond to the web tier's connection requests. Since Security Groups are stateful, the response traffic is already permitted.

- ⊗ Configure the security group for the database tier to allow inbound traffic on ports 443 and 1433 from the security group for the web tier.

Allowing inbound traffic on port 443 to the database server isn't required, SQL Server communicates on port 1433, opening other ports increases the attack surface and are unnecessary.

Overall Explanation:

To ensure secure communication between the web tier and the database tier, the security group for the web tier should allow inbound HTTPS traffic (port 443) from the internet (0.0.0.0/0). This enables public access to the web application. The security group for the database tier should only allow inbound traffic on port 1433 (default SQL Server port) from the security group associated with the web tier. This restricts database access to only the web tier instances, significantly improving security. Outbound rules from the web tier aren't required as security groups are stateful, allowing return traffic from established connections.

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Question 18

Not Attempted

Security



A company migrated millions of archival files to Amazon S3. A solutions architect needs to implement a solution that will encrypt all the archival data using a customer-provided key. The solution must encrypt existing unencrypted objects and future objects. Which solution will meet these requirements?

- ✔ Create a list of unencrypted objects by filtering an Amazon S3 Inventory report. Configure an S3 Batch Operations job to encrypt the objects from the list with server-side encryption with a customer-provided key (SSE-C). Configure the S3 default encryption feature to use server-side encryption with a customer-provided key (SSE-C). This option uses S3 Inventory to identify unencrypted objects and S3 Batch Operations to encrypt the existing unencrypted objects. S3 default encryption ensures future objects are encrypted using SSE-C.
- ⊗ Use S3 Storage Lens metrics to identify unencrypted S3 buckets. Configure the S3 default encryption feature to use server-side encryption with AWS KMS keys (SSE-KMS). S3 Storage Lens metrics identify unencrypted buckets, not individual objects. This approach also doesn't address encrypting the existing unencrypted objects and uses SSE-KMS instead of SSE-C.

- ⊗ Create a list of unencrypted objects by filtering the AWS usage report for Amazon S3. Configure an AWS Batch job to encrypt the objects from the list with server-side encryption with AWS KMS keys (SSE-KMS). Configure the S3 default encryption feature to use server-side encryption with AWS KMS keys (SSE-KMS).

AWS usage reports do not provide details on the encryption status of individual objects. Also, the question specifies using a customer-provided key, so SSE-KMS is incorrect.

- ⊗ Create a list of unencrypted objects by filtering the AWS usage report for Amazon S3. Configure the S3 default encryption feature to use a server-side encryption with a customer-provided key (SSE-C).

AWS usage reports do not provide details on the encryption status of individual objects. Also, this option doesn't address encrypting the existing unencrypted objects.

Overall Explanation:

The correct solution is to use S3 Inventory to identify unencrypted objects because it has an 'Encryption status' field and then use S3 Batch Operations to encrypt those existing objects with a customer-provided key (SSE-C). Finally, enable default encryption for the bucket with SSE-C to ensure all new objects are encrypted with the customer-provided key. This ensures both existing and future objects are encrypted as per the requirements.

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Question 19

Not Attempted

Networking



A company's website, hosted on AWS, is experiencing service disruptions due to outages at their current DNS provider. The company needs to quickly migrate to a more resilient managed DNS service on AWS. As a solutions architect, what is the MOST efficient way to migrate the DNS hosting service?

- ✔ Create an Amazon Route 53 public hosted zone for the domain name. Import the zone file containing the domain records hosted by the previous provider.

This is the correct approach. A public hosted zone in Route 53 is designed for publicly accessible domains and importing the zone file allows for a rapid migration of existing DNS records.

- ⊗ Create an Amazon Route 53 private hosted zone for the domain name. Import the zone file containing the domain records hosted by the previous provider.

A private hosted zone is used for internal DNS resolution within VPCs and would not make the website publicly accessible.

- ⊗ Create a Simple AD directory in AWS. Enable zone transfer between the DNS provider and AWS Directory Service for Microsoft Active Directory for the domain records.

Simple AD is not designed for public DNS hosting and zone transfers from the current provider are unlikely to be supported.

- ⊗ Create an Amazon Route 53 Resolver inbound endpoint in the VPC. Specify the IP addresses that the provider's DNS will forward DNS queries to. Configure the provider's DNS to forward DNS queries for the domain to the IP addresses that are specified in the inbound endpoint.

Route 53 Resolver inbound endpoints are used to resolve DNS queries originating from on-premises networks or other VPCs to resources within a VPC. This approach is unnecessarily complex for simply migrating public DNS records.

Overall Explanation:

The quickest and most appropriate solution is to create a public hosted zone in Amazon Route 53 and import the existing DNS records from the previous provider. This provides a resilient and managed DNS service on AWS for the company's public website. Private hosted zones are for internal DNS resolution and inbound resolvers are for on-premise DNS queries to AWS resources.

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Question 20

Not Attempted

Security



A company is developing an application on AWS that needs to connect to an Amazon RDS database. The company requires a solution to effectively manage the application configuration and securely store and retrieve database credentials, while minimizing administrative overhead.

- ✔ Use AWS AppConfig to store and manage the application configuration. Use AWS Secrets Manager to store and retrieve the credentials.

AWS AppConfig is designed for managing application configurations, and AWS Secrets Manager is purpose-built for securely storing and retrieving secrets like database credentials. This combination provides a managed solution with minimal administrative overhead.

- ⊗ Use AWS Lambda to store and manage the application configuration. Use AWS Systems Manager Parameter Store to store and retrieve the credentials.

Using Lambda to store application configuration is not its intended use and would require custom code and management overhead. While Parameter Store can store credentials, Secrets Manager is better suited and more secure for sensitive information.

- ⊗ Use an encrypted application configuration file. Store the file in Amazon S3 for the application configuration. Create another S3 file to store and retrieve the credentials.

While this solution is possible, managing encryption keys and access control for the S3 files adds significant administrative overhead compared to using managed services like AppConfig and Secrets Manager.

- ⊗ Use AWS AppConfig to store and manage the application configuration. Use Amazon RDS to store and retrieve the credentials.

Storing credentials directly in the RDS database is a security risk and is not a recommended practice. Secrets Manager is designed for secure credential storage and retrieval.

Overall Explanation:

The most efficient and secure solution is to use AWS AppConfig for application configuration management and AWS Secrets Manager for storing and retrieving database credentials. AppConfig provides a centralized and managed way to handle configurations, while Secrets Manager is specifically designed for secure credential management with features like rotation and access control. Options B, C, and D introduce unnecessary complexity and security risks compared to using these dedicated services.

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Question 21

Not Attempted

Security



A company needs to encrypt all of its application data in transit while communicating with an Amazon RDS MySQL DB instance to meet security requirements. A recent security audit revealed that encryption at rest is enabled using AWS Key Management Service (AWS KMS), but data in transit is not enabled. What should a solutions architect do to satisfy the security requirements?

- ⊗ Enable IAM database authentication on the database.

While IAM database authentication provides benefits, including encrypted network traffic, it doesn't fully address the need for ensuring SSL/TLS encryption. Further configuration with root certificates is still needed.

- ⊗ Provide self-signed certificates. Use the certificates in all connections to the RDS instance.

Using self-signed certificates is not recommended for production environments, as they can lead to trust issues and are more difficult to manage. AWS provides root certificates that should be used instead.

- ⊗ Take a snapshot of the RDS instance. Restore the snapshot to a new instance with encryption enabled.

This action is not related to enabling in-transit encryption. It addresses encryption at rest, which is already configured.

- ✓ Download AWS-provided root certificates. Provide the certificates in all connections to the RDS instance.

To encrypt data in transit to an RDS MySQL instance, you need to configure the database connection to use TLS/SSL, which requires using AWS-provided root certificates to establish a secure connection.

Overall Explanation:

To encrypt data in transit for an RDS MySQL instance, you need to configure TLS/SSL encryption on the database connections. This involves downloading the AWS-provided root certificates and configuring the clients to use them when connecting to the database. The RDS instance uses an AWS-provided certificate, so the application needs to be configured to trust that certificate.

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Question 22

Not Attempted

Networking



A company is designing a new web service that will run on Amazon EC2 instances behind an Elastic Load Balancing (ELB) load balancer. Many of the web service clients can only reach IP addresses authorized on their firewalls. What should a solutions architect recommend to meet the clients' needs?

- ✓ A Network Load Balancer with an associated Elastic IP address.

A Network Load Balancer (NLB) supports assigning Elastic IP addresses. These static IP addresses can then be whitelisted by the clients on their firewalls, ensuring they can reach the web service.

- ⊗ An Application Load Balancer with an associated Elastic IP address.

Application Load Balancers (ALB) do not support direct assignment of Elastic IP addresses. This option would not allow clients to whitelist specific IPs.

- ⊗ An A record in an Amazon Route 53 hosted zone pointing to an Elastic IP address.

While Route 53 can use A records, this option doesn't directly address associating a static IP with the load balancer itself, which is what the clients need to whitelist. An A record pointing to an Elastic IP would require an EC2 instance to hold that Elastic IP.

- ⊗ An EC2 instance with a public IP address running as a proxy in front of the load balancer.

While this could work, it introduces unnecessary complexity and management overhead. It's better to leverage the NLB's native Elastic IP support. This also creates a single point of failure and scaling bottleneck.

Overall Explanation:

Using a Network Load Balancer with an associated Elastic IP address is the most suitable solution. NLBs are designed for high throughput and low latency, and their support for static IP addresses allows clients to easily whitelist the web service in their firewalls.

Alternatives involve unnecessary complexity or don't directly solve the core problem.

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Question 23

Not Attempted

Security



A company has created a new AWS account. The account has the default settings and no changes have been made yet. The company is concerned about the security of the AWS account root user and wants to follow AWS best practices. What steps should the company take to enhance the security of the root user?

- ⊗ Create IAM users for daily administrative tasks. Disable the root user.

While creating IAM users for daily tasks is correct, disabling the root user is generally not recommended or fully possible. The root user is required for certain account-level tasks.

- ☒ Create IAM users for daily administrative tasks. Enable multi-factor authentication (MFA) on the root user.

This is the best option. Creating IAM users for day-to-day tasks limits the use of the root user. Enabling MFA on the root user provides an additional layer of security when the root user *must* be used.

- ☐ Generate an access key for the root user. Use the access key for daily administration tasks instead of the AWS Management Console.

Generating an access key for the root user and using it for daily tasks is a security risk and defeats the purpose of using IAM roles and users for granular permissions.

- ☐ Provide the root user credentials to the most senior solutions architect. Have the solutions architect use the root user for daily administration tasks.

Sharing root user credentials and using them for daily administration tasks is a major security risk and violates the principle of least privilege.

Overall Explanation:

The best practice is to create IAM users for daily tasks to avoid using the root user. Enabling Multi-Factor Authentication (MFA) on the root user's account is crucial as an additional layer of security when the root user *must* be used for specific tasks. The root user should not be used for regular administration. Segregation of duties dictates that root access must never be used for routine activities.

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Question 24

Not Attempted

Networking



A company is deploying an application that processes streaming data in near-real time using Amazon EC2 instances. The network architecture must be configured to minimize latency between nodes. Which combination of network solutions will meet these requirements? (Choose two.)

- ☒ Enable and configure enhanced networking on each EC2 instance.

Enhanced networking provides higher bandwidth, higher packet per second (PPS) performance, and consistently lower inter-instance latencies.

- ☐ Group the EC2 instances in separate accounts.

Grouping instances in separate accounts does not directly impact latency between nodes.

✔ Run the EC2 instances in a cluster placement group.

A cluster placement group is a logical grouping of instances within a single Availability Zone. This configuration is recommended for applications that need low network latency, high network throughput, or both. Cluster placement groups place instances close together physically, which reduces latency.

✘ Attach multiple elastic network interfaces to each EC2 instance.

Attaching multiple ENIs primarily increases network bandwidth, not necessarily reduces latency.

✘ Use Amazon Elastic Block Store (Amazon EBS) optimized instance types.

EBS-optimized instances improve throughput to EBS but do not directly reduce latency between instances.

Overall Explanation:

To achieve the lowest possible latency between EC2 instances, you should enable enhanced networking and run the instances in a cluster placement group. Enhanced networking provides better performance and lower latency, while a cluster placement group ensures the instances are physically located close together, minimizing network hops and latency.

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Question 25

Not Attempted

Storage



A financial services company needs to migrate over 100 TB of data from two on-premises data centers to AWS as part of a datacenter shutdown project. The existing data is unstructured, stored in an intricate directory structure with millions of small files, and uses SMB-based storage from various vendors. The company wants to minimize operational overhead and avoid application changes after the migration. Which solution BEST meets these requirements?

✘ Use AWS Direct Connect to migrate the data to Amazon S3.

While AWS Direct Connect provides a dedicated network connection, migrating directly to S3 might require application changes to access the data and could be complex given the existing SMB structure.

- ☐ Use AWS DataSync to migrate the data to Amazon FSx for Lustre.
Amazon FSx for Lustre is designed for high-performance computing and isn't the best fit for SMB-based storage and unstructured data. It does not natively support SMB file shares.
- ☒ Use AWS DataSync to migrate the data to Amazon FSx for Windows File Server.
AWS DataSync simplifies and accelerates data transfer between on-premises storage and AWS. FSx for Windows File Server supports SMB and preserves file system metadata, requiring minimal application changes. This option offers the least operational overhead.
- ☐ Use AWS Direct Connect to migrate the data on-premises file storage to an AWS Storage Gateway volume gateway.
While AWS Storage Gateway supports SMB and allows on-premises applications to access AWS storage, it introduces additional complexity and doesn't directly address the data migration and operational overhead concerns as effectively as DataSync.

Overall Explanation:

AWS DataSync is a data transfer service that simplifies, automates, and accelerates moving and replicating data between on-premises storage systems and AWS storage services. Given the company's desire to maintain SMB compatibility and minimize application changes, migrating the data using AWS DataSync to Amazon FSx for Windows File Server would be the most efficient and least operationally intensive solution. FSx for Windows File Server supports SMB and allows preserving file system metadata during the migration.

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Question 26

Not Attempted

[Monitoring and Observability](#)

A company uses an organization in AWS Organizations to manage AWS accounts that contain applications. The company sets up a dedicated monitoring member account in the organization. The company wants to query and visualize observability data across the accounts by using Amazon CloudWatch. Which solution will meet these requirements?

- ☒ Enable CloudWatch cross-account observability for the monitoring account. Deploy an AWS CloudFormation template provided by the monitoring account in each AWS account to share the data with the monitoring account.
CloudWatch cross-account observability allows you to query and visualize data from multiple AWS accounts in a single monitoring account. Deploying the CloudFormation template establishes the

necessary trust and permissions for data sharing.

- ⊗ Set up service control policies (SCPs) to provide access to CloudWatch in the monitoring account under the Organizations root organizational unit (OU).
SCPs control the maximum permissions available in member accounts. While they can restrict access, they do not inherently enable cross-account data querying and visualization.
- ⊗ Configure a new IAM user in the monitoring account. In each AWS account, configure an IAM policy to have access to query and visualize the CloudWatch data in the account. Attach the new IAM policy to the new IAM user.
This approach requires manual configuration of IAM policies in each account and is less scalable and manageable than using CloudWatch cross-account observability.
- ⊗ Create a new IAM user in the monitoring account. Create cross-account IAM policies in each AWS account. Attach the IAM policies to the new IAM user.
This approach is similar to Option C and also involves manual configuration of IAM policies in each account, making it less efficient than CloudWatch cross-account observability.

Overall Explanation:

CloudWatch cross-account observability is the most efficient and scalable solution for querying and visualizing data across multiple AWS accounts. It simplifies the process of data sharing and reduces the manual configuration required. Option A provides the correct configuration to setup CloudWatch cross-account observability.

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Question 27

Not Attempted

Security



A company website sells products and runs on EC2 instances in an Auto Scaling group behind an Application Load Balancer (ALB). An Amazon CloudFront distribution is used for caching, and AWS WAF protects against SQL injection. The ALB is the CloudFront origin. Security logs reveal a malicious external IP that needs to be blocked. What should a solutions architect do to protect the application?

- ⊗ Modify the network ACL on the CloudFront distribution to add a deny rule for the malicious IP address.

Network ACLs operate at the subnet level and are not the appropriate tool for blocking specific IPs at the application layer in front of CloudFront.

- ✔ Modify the configuration of AWS WAF to add an IP match condition to block the malicious IP address.

AWS WAF is designed to protect web applications from common exploits and can be configured to block specific IP addresses using IP match conditions.

- ✘ Modify the network ACL for the EC2 instances in the target groups behind the ALB to deny the malicious IP address.

Network ACLs operate at the subnet level. While they can block traffic, it is better to block the malicious traffic closer to the edge using WAF, preventing the traffic from reaching the ALB and EC2 instances.

- ✘ Modify the security groups for the EC2 instances in the target groups behind the ALB to deny the malicious IP address.

Security Groups are stateful and only support allow rules, not deny rules. Therefore, you cannot block traffic using security groups.

Overall Explanation:

The best approach is to use AWS WAF to block the malicious IP address. WAF is designed for this purpose and allows you to quickly and effectively block unwanted traffic at the application layer before it reaches your backend infrastructure. Modifying network ACLs is not as effective because it operates at a lower level (subnet) and doesn't provide the same level of application-specific filtering. Security groups can only allow traffic, not block it.

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Question 28

Not Attempted

Application Integration



A company is migrating a multi-tiered application from its on-premises data center to the AWS Cloud with the goal of enhancing application performance and modernizing its architecture. The application tiers communicate using RESTful services. Currently, the application experiences transaction drops when a tier becomes overloaded. As a solutions architect, you must design a solution that addresses these issues while ensuring operational efficiency. Which solution is the MOST suitable?

- ✔ Use Amazon API Gateway to route transactions to AWS Lambda functions for the application layer. Implement Amazon Simple Queue Service (Amazon SQS) as the communication layer between application services.

This solution leverages serverless architecture with Lambda for auto-scaling and SQS for decoupling, addressing both performance and modernization requirements. API Gateway handles RESTful service requests efficiently. This option is operationally efficient as it reduces the need for managing servers.

- ✘ Use Amazon CloudWatch metrics to analyze application performance history to identify peak utilization of servers during performance failures. Increase the size of the application server's Amazon EC2 instances to accommodate peak requirements.

While this may address performance issues, it doesn't modernize the application or improve operational efficiency. Scaling up EC2 instances might be costly and doesn't inherently address dropped transactions due to overloading.

- ✘ Use Amazon Simple Notification Service (Amazon SNS) to manage messaging between application servers running on Amazon EC2 in an Auto Scaling group. Use Amazon CloudWatch to monitor the SNS queue length and scale up and down as required.

SNS is primarily for pub/sub messaging, not for queuing and guaranteed delivery needed to prevent transaction drops. While Auto Scaling helps, it doesn't modernize the application and is less operationally efficient than serverless options.

- ✘ Use Amazon Simple Queue Service (Amazon SQS) to handle the messaging between application servers running on Amazon EC2 in an Auto Scaling group. Use Amazon CloudWatch to monitor the SQS queue length and scale up when communication failures are detected.

This solution uses EC2 instances, which is not modernizing the application, and also scales based on communication failures which is a reactive approach. This is less operationally efficient than Option A with API Gateway and Lambda.

Overall Explanation:

The best solution is to use Amazon API Gateway to route transactions to AWS Lambda functions, with Amazon SQS managing communication between application services. This approach modernizes the application using serverless technology (Lambda), improves performance with auto-scaling, and prevents transaction drops by decoupling the tiers with SQS. It is also the most operationally efficient, as it minimizes server management overhead.

Question 29

Not Attempted

Security



A company has an AWS Organizations setup with 10 accounts. They need to grant access to these accounts for thousands of employees, using their existing on-premises identity provider (IdP) for authentication. What is the MOST suitable AWS service that allows the company to federate access from their existing IdP to these AWS accounts?

- ☐ ☒ Create IAM users for all employees in each AWS account, connect these IAM users to the existing IdP, and configure federated authentication.
Creating IAM users in each account is not scalable for thousands of users and increases management overhead. Connecting IAM users to the IdP directly is complex to configure and manage.
- ☐ Set up AWS account root users with user email addresses and passwords synchronized from the existing IdP.
Using the root user is a security risk and not recommended for day-to-day access. Synchronizing credentials is also a complex task and does not align with best practices.
- ☒ Configure AWS IAM Identity Center (AWS Single Sign-On). Connect IAM Identity Center to the existing IdP. Provision users and groups from the IdP.
IAM Identity Center simplifies user management and provides centralized access control across multiple AWS accounts. It allows easy integration with existing IdPs and supports user and group provisioning, making it the most suitable choice for this scenario.
- ☐ Use AWS Resource Access Manager (RAM) to share access to the AWS accounts with users from the existing IdP.
AWS RAM is primarily used for sharing resources within your AWS organization or with other AWS accounts, not for federating identities from an external IdP.

Overall Explanation:

IAM Identity Center is the MOST suitable solution because it allows centralized management of users and permissions across multiple AWS accounts, seamlessly integrates with existing identity providers (IdPs), and simplifies the process of provisioning users and groups. IAM Identity Center redirects authentication to the existing IdP and grants access based on policies created in IAM Identity Center. Option A is not scalable.

Option B is a security risk and not recommended. Option D is incorrect as AWS RAM is used to share resources and not for federating identities.

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Question 30

Not Attempted

Security



A solutions architect is designing an AWS Identity and Access Management (IAM) authorization model for a company's AWS account. The company has designated five specific employees to have full access to AWS services and resources in the AWS account. The solutions architect has created an IAM user for each of the five designated employees and has created an IAM user group. Which solution will meet these requirements?

- ☐ Attach the AdministratorAccess resource-based policy to the IAM user group. Place each of the five designated employee IAM users in the IAM user group.
Resource-based policies are attached to resources, not IAM user groups. Therefore, this option is incorrect.
- ☐ Attach the SystemAdministrator identity-based policy to the IAM user group. Place each of the five designated employee IAM users in the IAM user group.
The SystemAdministrator policy provides limited access compared to AdministratorAccess. The requirement is to provide full access.
- ☒ Attach the AdministratorAccess identity-based policy to the IAM user group. Place each of the five designated employee IAM users in the IAM user group.
The AdministratorAccess policy grants full access to AWS services and resources. This policy attached to an IAM group containing the users will fulfill the requirement.
- ☐ Attach the SystemAdministrator resource-based policy to the IAM user group. Place each of the five designated employee IAM users in the IAM user group.
Resource-based policies are attached to resources, not IAM user groups. Additionally, SystemAdministrator policy does not grant full access.

Overall Explanation:

To provide full access to the five designated employees, the solutions architect should attach the AdministratorAccess identity-based policy to the IAM user group and place each of the five employee IAM users in the IAM user group. Identity-based policies are attached

to IAM identities (users, groups, roles), while resource-based policies are attached to AWS resources. AdministratorAccess provides full access, whereas SystemAdministrator offers limited privileges.

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Question 31

Not Attempted

Application Integration



A company's multi-tier payment processing application currently uses virtual machines (VMs). Communication between tiers is asynchronous via a third-party middleware guaranteeing exactly-once delivery. The company wants to migrate to a solution requiring the least infrastructure management while maintaining exactly-once delivery for application messaging. Which combination of actions achieves these goals? (Choose two.)

- ☒ Use AWS Lambda for the compute layers in the architecture.
AWS Lambda offers a serverless compute environment, minimizing infrastructure management. It automatically scales and manages the underlying resources, reducing operational overhead.
- ☐ Use Amazon EC2 instances for the compute layers in the architecture.
EC2 instances require manual management of the operating system, patching, scaling, and other infrastructure-related tasks, which does not meet the requirement for least infrastructure management.
- ☐ Use Amazon Simple Notification Service (Amazon SNS) as the messaging component between the compute layers.
Amazon SNS provides a pub/sub messaging service but does not guarantee exactly-once delivery. It's possible for messages to be lost or delivered multiple times.
- ☒ Use Amazon Simple Queue Service (Amazon SQS) FIFO queues as the messaging component between the compute layers.
Amazon SQS FIFO (First-In-First-Out) queues provide exactly-once processing and message ordering. This ensures each message is delivered exactly once and in the order it was sent, fulfilling the application's requirement.
- ☐ Use containers that are based on Amazon Elastic Kubernetes Service (Amazon EKS) for the compute layers in the architecture.
Amazon EKS, while offering container orchestration, requires more infrastructure management compared to Lambda. Managing the Kubernetes cluster, nodes, and networking adds complexity

and operational overhead.

Overall Explanation:

Using AWS Lambda for compute provides serverless operation, reducing management overhead. Amazon SQS FIFO queues ensure exactly-once message delivery, fulfilling the requirement. SNS doesn't provide exactly-once delivery, and EC2/EKS require more infrastructure management than Lambda.

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Question 32

Not Attempted

Storage



A company has a nightly batch processing routine that analyzes report files that an on-premises file system receives daily through SFTP. The company wants to move the solution to the AWS Cloud. The solution must be highly available and resilient. The solution also must minimize operational effort. Which solution meets these requirements?

- ☒ Deploy AWS Transfer for SFTP and an Amazon Elastic File System (Amazon EFS) file system for storage. Use an Amazon EC2 instance in an Auto Scaling group with a scheduled scaling policy to run the batch operation.

This option uses AWS Transfer for SFTP for a managed and highly available SFTP service and EFS for a shared, scalable file system. Using an EC2 instance in an Auto Scaling group with a scheduled scaling policy addresses high availability and resilience while also automating the batch processing.
- ☐ Deploy an Amazon EC2 instance that runs Linux and an SFTP service. Use an Amazon Elastic Block Store (Amazon EBS) volume for storage. Use an Auto Scaling group with the minimum number of instances and desired number of instances set to 1.

This option requires more operational effort because the company must manage the SFTP server on the EC2 instance. Also, EBS is not ideal for sharing data across multiple instances for high availability. Setting the desired number of instances to 1 negates the Auto Scaling benefits.
- ☐ Deploy an Amazon EC2 instance that runs Linux and an SFTP service. Use an Amazon Elastic File System (Amazon EFS) file system for storage. Use an Auto Scaling group with the minimum number of instances and desired number of instances set to 1.

This option requires more operational effort because the company must manage the SFTP server on the EC2 instance. Setting the desired number of instances to 1 negates the Auto Scaling benefits.

- ⊗ Deploy AWS Transfer for SFTP and an Amazon S3 bucket for storage. Modify the application to pull the batch files from Amazon S3 to an Amazon EC2 instance for processing. Use an EC2 instance in an Auto Scaling group with a scheduled scaling policy to run the batch operation.

While this option leverages managed services and Auto Scaling, it requires modifying the application to pull files from S3, which increases the operational effort. EFS support directly by Transfer Family reduces the complexity.

Overall Explanation:

Option A is the most suitable solution. It utilizes AWS Transfer for SFTP to provide a highly available and resilient SFTP service, and Amazon EFS for a shared, scalable file system. The use of an EC2 instance within an Auto Scaling group, configured with a scheduled scaling policy, ensures both high availability and resilience for the batch processing. This setup also minimizes operational overhead by leveraging managed services and automation, aligning with the requirement to minimize operational effort.

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Question 33

Not Attempted

Networking



A company has users all around the world accessing its HTTP-based application deployed on Amazon EC2 instances in multiple AWS Regions. The company wants to improve the availability and performance of the application. The company also wants to protect the application against common web exploits that may affect availability, compromise security, or consume excessive resources. Static IP addresses are required. What should a solutions architect recommend to accomplish this?

- ⊗ Put the EC2 instances behind Network Load Balancers (NLBs) in each Region. Deploy AWS WAF on the NLBs. Create an accelerator using AWS Global Accelerator and register the NLBs as endpoints.

NLBs do not natively support HTTP traffic or direct integration with AWS WAF. While Global Accelerator provides static IP addresses, the other components are not suitable for HTTP-based applications requiring WAF.

- ✔ Put the EC2 instances behind Application Load Balancers (ALBs) in each Region. Deploy AWS WAF on the ALBs. Create an accelerator using AWS Global Accelerator and register

the ALBs as endpoints.

This is the correct solution. ALBs support HTTP traffic and integrate with AWS WAF for protection against web exploits. AWS Global Accelerator provides static IP addresses and improves performance by routing traffic through the AWS global network.

- ⊗ Put the EC2 instances behind Network Load Balancers (NLBs) in each Region. Deploy AWS WAF on the NLBs. Create an Amazon CloudFront distribution with an origin that uses Amazon Route 53 latency-based routing to route requests to the NLBs.

NLBs are not designed for HTTP traffic. Also, while WAF can be deployed with CloudFront, NLB integration with CloudFront is not the most efficient or recommended approach. Static IPs are best provided via Global Accelerator.

- ⊗ Put the EC2 instances behind Application Load Balancers (ALBs) in each Region. Create an Amazon CloudFront distribution with an origin that uses Amazon Route 53 latency-based routing to route requests to the ALBs. Deploy AWS WAF on the CloudFront distribution.

While this solution uses ALBs and WAF, it lacks the static IP address requirement. CloudFront does not provide static IP addresses in the same manner as AWS Global Accelerator. CloudFront IP addresses are dynamic and can change.

Overall Explanation:

The correct answer is to use Application Load Balancers (ALBs) with AWS WAF and AWS Global Accelerator. ALBs handle HTTP traffic and integrate with AWS WAF for web exploit protection. AWS Global Accelerator provides static IP addresses, enhances availability, and improves application performance for a global user base. CloudFront's IPs are dynamic, and NLBs don't handle HTTP traffic.

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Question 34

Not Attempted

Databases



A company's data platform uses an Amazon Aurora MySQL database with multiple read replicas and DB instances across different Availability Zones. Users report database errors indicating too many connections. The company wants to decrease failover time by 20% when a read replica becomes the primary writer. Which solution will meet this requirement?

- ⊗ Switch from Aurora to Amazon RDS with Multi-AZ cluster deployment.

Switching to RDS with Multi-AZ does not directly address connection management or failover time reduction by 20%.

✔ Use Amazon RDS Proxy in front of the Aurora database.

Amazon RDS Proxy helps manage database connections, enabling connection pooling and reuse. This reduces the overhead of establishing new connections, leading to faster failover times and improved resilience. It also makes applications more resilient to database failures. When a primary database instance fails, RDS Proxy automatically connects to a standby DB instance while preserving application connections.

✘ Switch to Amazon DynamoDB with DynamoDB Accelerator (DAX) for read connections.

Switching to DynamoDB is a significant architectural change and doesn't directly reduce failover time for the existing Aurora database. DAX is for DynamoDB read caching, not Aurora.

✘ Switch to Amazon Redshift with relocation capability.

Switching to Redshift is a major architectural change and doesn't address the Aurora failover time requirement.

Overall Explanation:

Amazon RDS Proxy manages database connections by pooling and sharing them, improving scalability and resilience. During failover, RDS Proxy automatically connects to a standby instance while preserving application connections, thus reducing failover time. This directly addresses the requirement of reducing failover time by 20%. Therefore, the correct answer is B.

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Question 35

Not Attempted

Security



A company stores text files in Amazon S3. The text files include customer chat messages, date and time information, and customer personally identifiable information (PII). The company needs a solution to provide samples of the conversations to an external service provider for quality control. The external service provider needs to randomly pick sample conversations up to the most recent conversation. The company must not share the customer PII with the external service provider. The solution must scale when the number of customer conversations increases. Which solution will meet these requirements with the LEAST operational overhead?

- ✓ Create an Object Lambda Access Point. Create an AWS Lambda function that redacts the PII when the function reads the file. Instruct the external service provider to access the Object Lambda Access Point.

This is the correct answer. Object Lambda allows for on-the-fly data transformation (redaction of PII) before it's served to the external service provider, minimizing operational overhead and ensuring scalability.

- ✗ Create a batch process on an Amazon EC2 instance that regularly reads all new files, redacts the PII from the files, and writes the redacted files to a different S3 bucket. Instruct the external service provider to access the bucket that does not contain the PII.

This approach involves creating and managing an EC2 instance and a batch process, which adds operational overhead. It also introduces latency between new files being added and the redacted versions being available.

- ✗ Create a web application on an Amazon EC2 instance that presents a list of the files, redacts the PII from the files, and allows the external service provider to download new versions of the files that have the PII redacted.

This option requires managing a web application and EC2 instance, leading to significant operational overhead. It also relies on manual intervention for each file download.

- ✗ Create an Amazon DynamoDB table. Create an AWS Lambda function that reads only the data in the files that does not contain PII. Configure the Lambda function to store the non-PII data in the DynamoDB table when a new file is written to Amazon S3. Grant the external service provider access to the DynamoDB table.

Storing the data in DynamoDB introduces complexity and potential data consistency issues. It requires significant restructuring of the data and managing a DynamoDB table, making it operationally complex.

Overall Explanation:

Object Lambda Access Point with a Lambda function for PII redaction offers the least operational overhead by transforming the data on-the-fly when accessed. This solution is also highly scalable and ensures that the external service provider only receives the necessary data without PII. The other options involve managing EC2 instances, batch processes, web applications, or DynamoDB tables, all of which increase operational complexity.

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Question 36

Not Attempted

Compute



A company runs a legacy application on a single Amazon EC2 instance. The application code is immutable, and the system can only operate on one instance. A solutions architect needs to design a resilient solution to minimize recovery time in case of failure. Which of the following actions would be MOST suitable to meet these requirements?

- ☐ Enable termination protection for the EC2 instance.
Termination protection prevents accidental deletion of the instance but does not aid in recovery from failures.
- ☐ Configure the EC2 instance for Multi-AZ deployment.
The question explicitly states the application cannot run on more than one instance, making Multi-AZ deployment unsuitable.
- ☒ Create an Amazon CloudWatch alarm to recover the EC2 instance in case of failure.
CloudWatch alarms can be configured to automatically recover an EC2 instance upon detection of system failures, thus improving recovery time.
- ☐ Launch the EC2 instance with two Amazon Elastic Block Store (Amazon EBS) volumes that use RAID configurations for storage redundancy.
While RAID can improve data resilience, it does not address instance-level failures or improve recovery time as effectively as automated recovery.

Overall Explanation:

Using an Amazon CloudWatch alarm to recover the EC2 instance is the best option. It directly addresses the need to improve recovery time by automatically restarting the instance in case of failure. Termination protection prevents accidental deletion, Multi-AZ is not possible with a single instance application, and RAID configurations focus on data resilience rather than instance recovery.

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Question 37

Not Attempted

Compute



A company wants to deploy its containerized application workloads to a VPC across three Availability Zones. The company needs a solution that is highly available across Availability Zones. The solution must require minimal changes to the application and have the LEAST operational overhead. Which solution meets these requirements?

- ✔ Use Amazon Elastic Container Service (Amazon ECS). Configure Amazon ECS Service Auto Scaling to use target tracking scaling. Set the minimum capacity to 3. Set the task placement strategy type to spread with an Availability Zone attribute.

This is the correct answer. ECS with a spread placement strategy ensures high availability across Availability Zones and requires minimal operational overhead compared to EKS self-managed nodes or managing EC2 instances directly.

- ✘ Use Amazon Elastic Kubernetes Service (Amazon EKS) self-managed nodes. Configure Application Auto Scaling to use target tracking scaling. Set the minimum capacity to 3.

EKS self-managed nodes require you to manually install and configure the Kubernetes node components, such as kubelet, kube-proxy, and Docker, on your Amazon EC2 instances. You also need to manage the security group, IAM role, and subnet for your node group. This increases the operational overhead.

- ✘ Use Amazon EC2 Reserved Instances. Launch three EC2 instances in a spread placement group. Configure an Auto Scaling group to use target tracking scaling. Set the minimum capacity to 3.

This option involves managing EC2 instances directly, which includes patching, scaling, and other operational tasks, increasing the overall operational overhead.

- ✘ Use an AWS Lambda function. Configure the Lambda function to connect to a VPC. Configure Application Auto Scaling to use Lambda as a scalable target. Set the minimum capacity to 3.

Lambda is designed for event-driven workloads and may not be suitable for containerized application workloads without significant changes to the application architecture.

Overall Explanation:

Amazon ECS with the spread placement strategy is the best choice because it minimizes operational overhead while ensuring high availability across multiple Availability Zones for containerized applications. EKS with self-managed nodes and EC2 instances require much more management overhead. Lambda is not suitable for containerized applications.

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Question 38

Not Attempted

Storage



A media company stores movies in Amazon S3. Each movie is stored in a single video file that ranges from 1 GB to 10 GB in size. The company must be able to provide the streaming content of a movie within 5 minutes of a user purchase. There is higher demand for movies that are less than 20 years old than for movies that are more than 20 years old. The company wants to minimize hosting service costs based on demand. Which solution will meet these requirements?

- ☐ Store all media content in Amazon S3. Use S3 Lifecycle policies to move media data into the Infrequent Access tier when the demand for a movie decreases.

S3 Lifecycle policies cannot be triggered based on real-time demand patterns. They are based on object age or other fixed criteria. Demand for an older movie could suddenly increase.

- ☒ Store newer movie video files in S3 Standard. Store older movie video files in S3 Standard-infrequent Access (S3 Standard-IA). When a user orders an older movie, retrieve the video file by using standard retrieval.

This is the most cost-effective solution that meets the requirements. S3 Standard-IA provides low latency access needed for streaming within 5 minutes, while also being more cost-effective for less frequently accessed older movies.

- ☐ Store newer movie video files in S3 Intelligent-Tiering. Store older movie video files in S3 Glacier Flexible Retrieval. When a user orders an older movie, retrieve the video file by using expedited retrieval.

While Glacier Flexible Retrieval offers expedited retrieval, expedited retrieval is not guaranteed within 1-5 minutes for files larger than 250 MB. Additionally, it's designed for occasional urgent requests, not regular streaming. Intelligent-Tiering is good, but the older movies retrieval is not the best.

- ☐ Store newer movie video files in S3 Standard. Store older movie video files in S3 Glacier Flexible Retrieval. When a user orders an older movie, retrieve the video file by using bulk retrieval.

Bulk retrieval from Glacier Flexible Retrieval takes hours, which does not meet the 5-minute requirement for streaming content.

Overall Explanation:

Storing newer movies in S3 Standard and older movies in S3 Standard-IA balances cost and accessibility. S3 Standard-IA offers rapid access with lower storage costs, meeting the 5-minute streaming requirement and minimizing expenses based on demand. Other options

either don't guarantee timely retrieval or rely on lifecycle policies that can't adapt to fluctuating demand.

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Question 39

Not Attempted

Networking



A company receives 10 TB of instrumentation data each day from several machines located at a single factory. The data consists of JSON files stored on a storage area network (SAN) in an on-premises data center located within the factory. The company wants to send this data to Amazon S3 where it can be accessed by several additional systems that provide critical near-real-time analytics. A secure transfer is important because the data is considered sensitive. Which solution offers the MOST reliable data transfer?

- ☐ AWS DataSync over public internet
Transferring data over the public internet introduces potential security risks and performance variability.
- ☒ AWS DataSync over AWS Direct Connect
AWS DataSync is designed for efficient data transfer and AWS Direct Connect provides a dedicated, private connection for reliable data transfer.
- ☐ AWS Database Migration Service (AWS DMS) over public internet
AWS DMS is primarily focused on database migration, not general file transfer, and the public internet is less reliable.
- ☐ AWS Database Migration Service (AWS DMS) over AWS Direct Connect
AWS DMS is primarily focused on database migration, not general file transfer, and is not as efficient as DataSync for transferring JSON files to S3.

Overall Explanation:

AWS DataSync is designed for large-scale, high-speed data transfer between on-premises storage and S3. Using AWS Direct Connect provides a dedicated, private connection for reliable and consistent data transfer. DMS is not optimized for transferring large volumes of JSON files. Furthermore, transferring data over the public internet is less secure and reliable than using a dedicated connection.

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Question 40

Not Attempted

Storage



A solutions architect needs to design the architecture for an application that a vendor provides as a Docker container image. The container requires 50 GB of storage available for temporary files. The infrastructure must be serverless and minimize operational overhead. Which solution meets these requirements?

- ☐ Create an AWS Lambda function that uses the Docker container image with an Amazon S3 mounted volume that has more than 50 GB of space.
Lambda is not ideal for long-running containerized applications, and S3 is not designed for temporary file storage.
- ☐ Create an AWS Lambda function that uses the Docker container image with an Amazon Elastic Block Store (Amazon EBS) volume that has more than 50 GB of space.
Lambda is not ideal for managing large volumes of data. Also, attaching EBS volumes to Lambda functions is not a standard or supported practice.
- ☒ Create an Amazon Elastic Container Service (Amazon ECS) cluster that uses the AWS Fargate launch type. Create a task definition for the container image with an Amazon Elastic File System (Amazon EFS) volume. Create a service with that task definition.
ECS with Fargate provides a serverless container execution environment. EFS allows for shared storage that is suitable for the container and meets the 50GB+ storage requirement, all with minimal operational overhead.
- ☐ Create an Amazon Elastic Container Service (Amazon ECS) cluster that uses the Amazon EC2 launch type with an Amazon Elastic Block Store (Amazon EBS) volume that has more than 50 GB of space. Create a task definition for the container image. Create a service with that task definition.
EC2 launch type requires you to manage the underlying EC2 instances, increasing operational overhead. While EBS can provide the storage, it makes the setup not serverless.

Overall Explanation:

The combination of ECS with Fargate and EFS (option C) provides a serverless solution that can run Docker containers and meet the storage requirements, all while minimizing

operational overhead. You don't need to manage any servers, and the storage will automatically scale as needed. This makes it the best fit for the given requirements.

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Question 41

Not Attempted

Security



A company needs to enable its users to authenticate to the AWS Management Console using their existing on-premises LDAP directory. The LDAP directory service is not compatible with SAML. What is the MOST suitable solution to meet these requirements?

- ☐ Enable AWS IAM Identity Center (AWS Single Sign-On) between AWS and the on-premises LDAP.
AWS IAM Identity Center requires compatibility with SAML or other supported federation methods. It does not natively support direct integration with non-SAML LDAP.
- ☐ Create an IAM policy that uses AWS credentials, and integrate the policy into LDAP.
IAM policies do not directly integrate with LDAP for authentication purposes. IAM policies define permissions, not authentication mechanisms.
- ☐ Set up a process that rotates the IAM credentials whenever LDAP credentials are updated.
This is a complicated and insecure solution. Rotating IAM credentials based on LDAP updates is not a standard or recommended practice.
- ☒ Develop an on-premises custom identity broker application or process that uses AWS Security Token Service (AWS STS) to get short-lived credentials.
A custom identity broker can authenticate users against LDAP and then use AWS STS to obtain temporary, short-lived credentials for accessing AWS resources. This allows authentication against LDAP without SAML, meeting the stated requirements.

Overall Explanation:

Since the on-premises LDAP is not SAML compatible, AWS IAM Identity Center (SSO) cannot be directly integrated. The best approach is to build a custom identity broker that authenticates against the LDAP directory and then uses AWS STS to retrieve temporary AWS credentials for the user. Options A, B and C are not viable solutions as they do not

address the SAML incompatibility or provide a secure method for authentication. Therefore, option D is the correct answer.

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Question 42

Not Attempted

Storage



A company stores multiple Amazon Machine Images (AMIs) in an AWS account to launch its Amazon EC2 instances. The AMIs contain critical data and configurations that are necessary for the company's operations. The company wants to implement a solution that will recover accidentally deleted AMIs quickly and efficiently. Which solution will meet these requirements with the LEAST operational overhead?

- ☐ Create Amazon Elastic Block Store (Amazon EBS) snapshots of the AMIs. Store the snapshots in a separate AWS account.

Creating EBS snapshots and storing them in a separate account adds operational overhead for management and incurs costs for storage and transfer.

- ☐ Copy all AMIs to another AWS account periodically.

Copying AMIs to another account involves significant overhead in terms of time, storage, and management. It's not the most efficient approach for quick recovery.

- ☒ Create a retention rule in Recycle Bin.

Recycle Bin allows you to restore accidentally deleted AMIs within a specified retention period with minimal operational overhead. This is the most efficient solution.

- ☐ Upload the AMIs to an Amazon S3 bucket that has Cross-Region Replication.

Uploading AMIs to S3 requires converting them to a different format and adds complexity. Cross-Region Replication adds cost and may not be as straightforward as Recycle Bin for direct AMI recovery.

Overall Explanation:

Recycle Bin is a data recovery feature that enables you to restore accidentally deleted Amazon EBS snapshots and EBS-backed AMIs. When using Recycle Bin, if your resources are deleted, they are retained in the Recycle Bin for a time period that you specify before being permanently deleted. You can restore a resource from the Recycle Bin at any time before its retention period expires. This solution has the least operational overhead, as you

do not need to create, copy, or upload any additional resources. You can also manage tags and permissions for AMIs in the Recycle Bin. AMIs in the Recycle Bin do not incur any additional charges.

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Question 43

Not Attempted

Cost Optimization



A company needs to migrate 150 TB of archived image data from its on-premises data center to AWS within one month. Their current network connection allows for a maximum upload speed of 100 Mbps during nighttime hours only. Which method is the MOST cost-effective way to achieve this data migration within the given timeframe?

☐ Use AWS Snowmobile to ship the data to AWS.

AWS Snowmobile is designed for very large data transfers (exabytes) and has a higher upfront cost compared to Snowball. It is not the most cost-effective option for 150 TB of data.

☒ Order multiple AWS Snowball devices to ship the data to AWS.

AWS Snowball is a cost-effective data transfer service for moving large amounts of data into and out of AWS. Given the amount of data and the time constraint, using multiple Snowball devices is the most efficient and cost-effective solution.

☐ Enable Amazon S3 Transfer Acceleration and securely upload the data.

While S3 Transfer Acceleration can improve upload speeds, the limited network bandwidth (100 Mbps) and the large data volume will make it a very time-consuming and potentially expensive option.

☐ Create an Amazon S3 VPC endpoint and establish a VPN to upload the data.

Creating an S3 VPC endpoint and a VPN connection doesn't address the bottleneck of the limited network bandwidth. This method would still take a significant amount of time to transfer the data, making it a less effective solution.

Overall Explanation:

Given the 150 TB data volume and a one-month deadline, using multiple AWS Snowball devices offers the most cost-effective solution. Snowball eliminates the dependency on the company's limited network bandwidth and allows for faster data transfer to AWS.

Snowmobile is overkill for this data volume and significantly more expensive. S3 Transfer

Overall Explanation:

Option B offers the best balance of minimal changes and required functionality. Aurora's point-in-time recovery simplifies management compared to RDS, and the separation of tiers into public (web) and private (application and database) subnets enhances security.

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Question 45

Not Attempted

Application Integration



A development team is collaborating with another company to create an integrated product. The other company needs to access an Amazon Simple Queue Service (Amazon SQS) queue that is contained in the development team's account. The other company wants to poll the queue without giving up its own account permissions to do so. How should a solutions architect provide access to the SQS queue?

- ☐ Create an instance profile that provides the other company access to the SQS queue.
Instance profiles are used to grant permissions to EC2 instances, not for granting access to other AWS services like SQS queues across accounts. This is not the correct approach.
- ☐ Create an IAM policy that provides the other company access to the SQS queue.
IAM policies are applied to IAM users, groups, or roles within the same AWS account. They cannot directly grant access to resources in other AWS accounts without additional mechanisms. This option doesn't fulfill the requirement of not relinquishing the other company's own account permissions.
- ☒ Create an SQS access policy that provides the other company access to the SQS queue.
SQS access policies allow you to grant cross-account access to SQS resources. You can specify the necessary permissions in the policy and attach it directly to the SQS queue. This allows the other company's AWS account to poll the queue without compromising its own account permissions.
- ☐ Create an Amazon Simple Notification Service (Amazon SNS) access policy that provides the other company access to the SQS queue.
Amazon SNS access policies are used to manage access to SNS topics, not SQS queues. This option is incorrect.

Overall Explanation:

The correct solution is to create an SQS access policy. This allows granting fine-grained, cross-account access to the SQS queue without the other company needing to modify its own IAM permissions. SQS access policies are resource-based policies that directly control who can access the queue and what actions they can perform.

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Question 46

Not Attempted

Security



A company's developers need a secure and cost-effective way to gain SSH access to the company's Amazon EC2 instances, which run the latest version of Amazon Linux. The developers work both remotely and in the corporate office. The EC2 instances are hosted in a VPC private subnet and access the internet through a NAT gateway deployed in a public subnet. Which solution BEST meets these requirements?

- ☐ Create a bastion host in the same subnet as the EC2 instances. Grant the `ec2:CreateVpnConnection` IAM permission to the developers. Install EC2 Instance Connect so that the developers can connect to the EC2 instances.
Creating a bastion host in the same subnet as the EC2 instances is not a secure architecture. Granting `ec2:CreateVpnConnection` to developers does not address SSH access. EC2 Instance Connect could work, but is not the most cost effective solution.
- ☐ Create an AWS Site-to-Site VPN connection between the corporate network and the VPC. Instruct the developers to use the Site-to-Site VPN connection to access the EC2 instances when the developers are on the corporate network. Instruct the developers to set up another VPN connection for access when they work remotely.
This solution involves setting up and managing VPN connections, which is a complex and expensive approach.
- ☐ Create a bastion host in the public subnet of the VPC. Configure the security groups and SSH keys of the bastion host to only allow connections and SSH authentication from the developers' corporate and remote networks. Instruct the developers to connect through the bastion host by using SSH to reach the EC2 instances.
While a bastion host provides SSH access, it introduces instance running costs and requires ongoing maintenance, making it less cost-effective than other solutions.
- ☒ Attach the `AmazonSSMManagedInstanceCore` IAM policy to an IAM role that is associated with the EC2 instances. Instruct the developers to use AWS Systems Manager

Session Manager to access the EC2 instances.

AWS Systems Manager Session Manager offers secure and cost-effective SSH access without the need for SSH keys or bastion hosts. It leverages IAM policies for access control and provides encrypted connections by default.

Overall Explanation:

AWS Systems Manager Session Manager is the most cost-effective and secure solution. It avoids the complexity and expense of VPNs and bastion hosts, and it eliminates the need to manage SSH keys. By attaching the AmazonSSMManagedInstanceCore IAM policy to the EC2 instances' IAM role, and granting developers the necessary permissions, they can securely access the instances through the AWS Management Console, CLI, or SDKs.

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Question 47

Not Attempted

Storage



A pharmaceutical company is developing a new drug. The volume of data that the company generates has grown exponentially over the past few months. The company's researchers regularly require a subset of the entire dataset to be immediately available with minimal lag. However, the entire dataset does not need to be accessed on a daily basis. All the data currently resides in on-premises storage arrays, and the company wants to reduce ongoing capital expenses. Which storage solution should a solutions architect recommend to meet these requirements?

- ☐ Run AWS DataSync as a scheduled cron job to migrate the data to an Amazon S3 bucket on an ongoing basis.

DataSync, while useful for migration, doesn't address the requirement of having a subset of data readily available with minimal lag. This option also involves regularly transferring the data even when not needed, increasing costs.

- ☐ Deploy an AWS Storage Gateway file gateway with an Amazon S3 bucket as the target storage. Migrate the data to the Storage Gateway appliance.

A file gateway would require storing the entire dataset in S3, which may not be cost-effective if only a subset needs to be accessed regularly. It also introduces potential latency when accessing data directly from S3.

- ✓ Deploy an AWS Storage Gateway volume gateway with cached volumes with an Amazon S3 bucket as the target storage. Migrate the data to the Storage Gateway appliance.

This solution uses cached volumes, allowing frequently accessed data to be stored locally for low-latency access while storing the entire dataset in S3 for cost-effectiveness. The on-premises Storage Gateway provides quick access to the frequently needed data.

- ✗ Configure an AWS Site-to-Site VPN connection from the on-premises environment to AWS. Migrate data to an Amazon Elastic File System (Amazon EFS) file system.

Using EFS directly over a VPN connection might introduce significant latency, especially if the connection is not highly performant. Also, keeping the entire dataset in EFS can be more expensive than using S3 for archival storage and caching a subset locally.

Overall Explanation:

The best option is to deploy an AWS Storage Gateway volume gateway with cached volumes. This allows the company to keep frequently accessed data locally for minimal lag, while storing the less frequently accessed data in Amazon S3 to reduce ongoing capital expenses. Storage arrays are also known as disk array, and AWS Storage Gateway volume gateway should be used to integrate the on-premise storage array with cloud storage.

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Question 48

Not Attempted

Databases



A company has a business-critical application that runs on Amazon EC2 instances. The application stores data in an Amazon DynamoDB table. The company must be able to revert the table to any point within the last 24 hours. Which solution meets these requirements with the LEAST operational overhead?

- ✓ Configure point-in-time recovery for the table.

Point-in-time recovery protects DynamoDB tables from accidental write or delete operations. It allows restoring to any point in time during the last 35 days (with granularity of minutes). This eliminates the need for manual backup creation, maintenance, and scheduling, which results in minimal operational overhead.

- ✗ Use AWS Backup for the table.

AWS Backup requires configuration and scheduling of backups, which introduces operational overhead. While it's a valid backup solution, it's not the least overhead approach for restoring to any point within a 24-hour window.

- ⊗ Use an AWS Lambda function to make an on-demand backup of the table every hour.
Using a Lambda function to make hourly backups requires writing, deploying, and maintaining the Lambda function, as well as managing the storage of the backups. This adds significant operational overhead compared to point-in-time recovery.
- ⊗ Turn on streams on the table to capture a log of all changes to the table in the last 24 hours. Store a copy of the stream in an Amazon S3 bucket.

DynamoDB Streams capture changes but rebuilding the table from streams to a specific point in time requires custom code and significant effort, making this approach highly complex and not the least overhead option. Also, DynamoDB Streams have a default retention period of 24 hours, copying to S3 would increase operational complexity.

Overall Explanation:

Point-in-time recovery for DynamoDB offers the least operational overhead because it is a built-in feature that automatically handles incremental backups and allows restoration to any point in time within the defined recovery window. This eliminates the need for manual backup creation, scheduling, and management, reducing the overall operational burden. Options B, C, and D all require more configuration and maintenance.

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Question 49

Not Attempted

Cost Optimization



A company hosts an application that uploads files to an Amazon S3 bucket. After uploading, the files are processed to extract metadata, which takes less than 5 seconds. The volume and frequency of uploads fluctuate from a few files hourly to hundreds of concurrent uploads. A solutions architect needs to design a cost-effective architecture that meets these requirements. Which architecture should the solutions architect recommend?

- ⊗ Configure AWS CloudTrail trails to log S3 API calls. Use AWS AppSync to process the files.

AWS CloudTrail is for auditing API calls and AWS AppSync is not ideal for simple file processing. It's more suitable for real-time data synchronization.

- ✔ Configure an object-created event notification within the S3 bucket to invoke an AWS Lambda function to process the files.

This is the most cost-effective and appropriate solution. S3 event notifications directly trigger a Lambda function upon file upload, allowing for immediate metadata extraction.

- ✘ Configure Amazon Kinesis Data Streams to process and send data to Amazon S3. Invoke an AWS Lambda function to process the files.

Kinesis Data Streams is designed for real-time data streaming and analytics. It is an overkill and not cost-effective for this simple file processing task.

- ✘ Configure an Amazon Simple Notification Service (Amazon SNS) topic to process the files uploaded to Amazon S3. Invoke an AWS Lambda function to process the files.

While SNS can trigger actions based on S3 events, it introduces an unnecessary layer of complexity compared to directly invoking Lambda from S3 event notifications.

Overall Explanation:

The most cost-effective solution is to use S3 event notifications to trigger an AWS Lambda function. This directly addresses the requirement of processing files upon upload without the overhead of services like Kinesis or the inappropriateness of CloudTrail and AppSync for this specific scenario. S3 event notifications are designed for this type of event-driven processing, and Lambda provides a serverless, pay-per-execution compute environment perfectly suited for the quick metadata extraction.

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Question 50

Not Attempted

Application Integration



A company needs to configure a real-time data ingestion architecture for its application. The company needs an API endpoint, a process that transforms data as the data is streamed, and a storage solution for the data. Which solution will meet these requirements with the LEAST operational overhead?

- ✘ Deploy an Amazon EC2 instance to host an API that sends data to an Amazon Kinesis data stream. Create an Amazon Kinesis Data Firehose delivery stream that uses the Kinesis data stream as a data source. Use AWS Lambda functions to transform the data. Use the Kinesis Data Firehose delivery stream to send the data to Amazon S3.

Using an EC2 instance for the API adds operational overhead. API Gateway provides a serverless API solution.

- ⊗ Deploy an Amazon EC2 instance to host an API that sends data to AWS Glue. Stop source/destination checking on the EC2 instance. Use AWS Glue to transform the data and to send the data to Amazon S3.

Using an EC2 instance for the API adds operational overhead. AWS Glue is primarily for batch processing, not real-time ingestion, and it doesn't directly receive data from APIs.

- ✔ Configure an Amazon API Gateway API to send data to an Amazon Kinesis data stream. Create an Amazon Kinesis Data Firehose delivery stream that uses the Kinesis data stream as a data source. Use AWS Lambda functions to transform the data. Use the Kinesis Data Firehose delivery stream to send the data to Amazon S3.

This option utilizes fully managed services (API Gateway, Kinesis Data Streams, Lambda, and Kinesis Data Firehose) minimizing operational overhead. API Gateway handles the API, Kinesis Data Streams ingests data, Lambda transforms it, and Firehose delivers to S3.

- ⊗ Configure an Amazon API Gateway API to send data to AWS Glue. Use AWS Lambda functions to transform the data. Use AWS Glue to send the data to Amazon S3.

AWS Glue is designed for batch processing, not real-time data ingestion. Kinesis Data Streams and Firehose are better suited for real-time scenarios.

Overall Explanation:

Option C provides the least operational overhead by leveraging Amazon API Gateway for the API, Amazon Kinesis Data Streams for real-time data ingestion, AWS Lambda for data transformation, and Amazon Kinesis Data Firehose to deliver the transformed data to Amazon S3. This combination of services avoids the need for managing EC2 instances or using AWS Glue for real-time processing.

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Question 51

Not Attempted

Cost Optimization



A company's application is deployed on Amazon EC2 instances and uses AWS Lambda functions for an event-driven architecture. The company uses nonproduction development environments in a different AWS account to test new features before deploying them to production. The production instances show constant usage because of customers in different

time zones. The company uses nonproduction instances only during business hours on weekdays and not on weekends. The company wants to optimize costs to run its application on AWS. Which solution will meet these requirements MOST cost-effectively?

- ⊗ Use On-Demand Instances for the production instances. Use Dedicated Hosts for the nonproduction instances on weekends only.

Using On-Demand Instances for production would be expensive given the constant usage. Dedicated Hosts are unnecessary and expensive for nonproduction instances used only during business hours.

- ⊗ Use Reserved Instances for the production instances and the nonproduction instances. Shut down the nonproduction instances when not in use.

While Reserved Instances are a good option for production, they are not ideal for nonproduction environments used only during business hours, as you would be paying for unused capacity during off-peak times. Shutting down the non-production will only partially offset cost.

- ✔ Use Compute Savings Plans for the production instances. Use On-Demand Instances for the nonproduction instances. Shut down the nonproduction instances when not in use.

Compute Savings Plans provide cost savings for consistent compute usage, making them suitable for production instances. Using On-Demand Instances for nonproduction instances and shutting them down when not in use optimizes costs, as you only pay for the time they are running.

- ⊗ Use Dedicated Hosts for the production instances. Use EC2 Instance Savings Plans for the nonproduction instances.

Dedicated Hosts are generally an expensive option and are not required for this scenario. Instance Savings Plans are also not optimal for non-production instances that can be shut down, as it commits to specific instance types regardless of utilization.

Overall Explanation:

Compute Savings Plans are ideal for production instances with consistent usage, providing cost savings compared to On-Demand. Using On-Demand Instances for nonproduction environments combined with shutting them down when not in use provides the most cost-effective solution by only paying for resources when they are actively being used.

Dedicated Hosts are not cost-effective or necessary. Reserved Instances are less efficient than Savings Plans for consistent workloads and are less flexible than On-Demand for variable workloads.

Question 52

Not Attempted

Databases



A company needs to migrate its on-premises Oracle database to Amazon Aurora PostgreSQL for analysis. The company has a Site-to-Site VPN connection to AWS. During the migration, they need to capture all ongoing changes to the Oracle database and replicate them to Aurora PostgreSQL. Which solution meets these requirements?

- ☐ Use the AWS Schema Conversion Tool (AWS SCT) to convert the Oracle schema to Aurora PostgreSQL schema. Use the AWS Database Migration Service (AWS DMS) full-load migration task to migrate the data.

This option performs the schema conversion and initial data load but does not address the requirement of capturing ongoing changes during the migration.

- ☐ Use AWS DataSync to migrate the data to an Amazon S3 bucket. Import the S3 data to Aurora PostgreSQL by using the Aurora PostgreSQL `aws_s3` extension.

This option facilitates data transfer to S3 but doesn't directly convert the schema or capture ongoing changes from the source Oracle database.

- ☒ Use the AWS Schema Conversion Tool (AWS SCT) to convert the Oracle schema to Aurora PostgreSQL schema. Use AWS Database Migration Service (AWS DMS) to migrate the existing data and replicate the ongoing changes.

This option correctly uses AWS SCT for schema conversion and AWS DMS for both initial data migration and ongoing change replication, meeting the requirements.

- ☐ Use an AWS Snowball device to migrate the data to an Amazon S3 bucket. Import the S3 data to Aurora PostgreSQL by using the Aurora PostgreSQL `aws_s3` extension.

This option uses Snowball for initial data transfer, which is unnecessary given the VPN connection, and it does not address schema conversion or ongoing change replication.

Overall Explanation:

The best solution is to use AWS SCT to convert the Oracle schema to Aurora PostgreSQL and AWS DMS to migrate existing data and replicate ongoing changes. This covers both the initial data migration and the capture of changes during the process. Options A, B, and D do not fully satisfy the requirement of capturing changes during migration.

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Question 53

Not Attempted

Compute



A company has developed an application using Docker containers and intends to deploy it on the AWS Cloud. The company prefers a managed service to host the application, ensuring automatic scaling based on demand and minimal operational overhead. Which two solutions best meet these requirements?

- ☒ Use Amazon Elastic Container Service (Amazon ECS) with AWS Fargate.
ECS with Fargate allows you to run containers without managing the underlying infrastructure. Fargate provides on-demand, right-sized compute capacity to run your containers, automatically scaling to match your application's needs.
- ☒ Use Amazon Elastic Kubernetes Service (Amazon EKS) with AWS Fargate.
EKS with Fargate offers a managed Kubernetes service, enabling you to run containerized applications without managing the Kubernetes control plane or worker nodes. Fargate handles the underlying infrastructure, providing automatic scaling based on demand.
- ☐ Provision an Amazon API Gateway API. Connect the API to AWS Lambda to run the containers.
While Lambda can run container images, API Gateway and Lambda are better suited for microservices and event-driven functions. Directly managing and scaling complex containerized applications with this approach is not ideal and would result in more operational overhead compared to ECS/EKS with Fargate.
- ☐ Use Amazon Elastic Container Service (Amazon ECS) with Amazon EC2 worker nodes.
Using ECS with EC2 worker nodes requires managing the EC2 instances, including scaling, patching, and maintenance. This results in increased operational overhead, which the company wants to avoid.
- ☐ Use Amazon Elastic Kubernetes Service (Amazon EKS) with Amazon EC2 worker nodes.
Similar to ECS with EC2, EKS with EC2 worker nodes requires managing the EC2 instances, increasing operational overhead. The company seeks a managed solution with minimal infrastructure management.

Overall Explanation:

The best solutions are ECS with Fargate and EKS with Fargate. Both provide managed container orchestration without requiring the company to manage the underlying infrastructure. Fargate handles the scaling and provisioning of compute resources, aligning with the requirement for automatic scaling and minimal operational overhead. API

Gateway and Lambda are not designed for directly managing and scaling Docker containers. Options involving EC2 worker nodes increase operational overhead, which is undesirable.

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Question 54

Not Attempted

Cost Optimization



An e-commerce company hosts its website on Amazon EC2 instances across multiple Availability Zones. They anticipate sudden traffic spikes during a seasonal online sale and want to manage these increases cost-effectively. Which solution is the MOST suitable?

- ☐ Create an Auto Scaling group large enough to handle peak traffic. Stop half the EC2 instances. Configure the group to use stopped instances to scale out when traffic increases.

This approach involves stopping instances, which can lead to delays during scaling as stopped instances take longer to become available compared to launching new ones. It's also less cost-effective to pay for instances even when they are stopped.

- ☐ Create an Auto Scaling group for the website. Set the minimum size of the Auto Scaling group so that it can handle high traffic volumes without the need to scale out.

While this ensures capacity, it's not cost-effective. The company would be paying for the resources required for peak traffic volumes even when the demand is low.

- ☐ Use Amazon CloudFront and Amazon ElastiCache to cache dynamic content with an Auto Scaling group set as the origin. Configure the Auto Scaling group with the instances necessary to populate CloudFront and ElastiCache. Scale in after the cache is fully populated.

Using CloudFront and ElastiCache can significantly improve performance and reduce load on the EC2 instances. However, scaling in after cache population might lead to performance issues during peak times if the cache misses increase. It is also more complex and potentially more expensive than a simple scaling solution.

- ☒ Configure an Auto Scaling group to scale out as traffic increases. Create a launch template to start new instances from a preconfigured Amazon Machine Image (AMI).

This is the most cost-effective and efficient solution. It allows the company to only pay for the instances they need, scaling up and down based on demand. Using a launch template and preconfigured AMI ensures instances are quickly provisioned with the necessary configurations.

Overall Explanation:

The most cost-effective approach is to configure an Auto Scaling group that scales out as needed based on traffic demands. Using a launch template and pre-configured AMI streamlines the scaling process, ensuring that new instances can be brought online quickly and efficiently. Options A and B are less cost-effective as they involve either paying for unused capacity or incurring delays in scaling. While Option C leverages caching, it is more complex and might not be as cost-effective for simply scaling EC2 instances.

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Question 55

Not Attempted

Security



A solutions architect needs to implement an automated solution to enforce a company's security policy. The policy prohibits security group rules that allow SSH access (port 22) from 0.0.0.0/0 (anywhere). The company must be notified immediately if the policy is violated. The solution must minimize operational overhead. Which approach is the MOST efficient?

- ☐ Create an AWS Lambda function that periodically scans security groups for SSH rules open to 0.0.0.0/0 and sends a notification upon detection.
While this approach works, it requires custom code development and maintenance, leading to higher operational overhead.
- ☒ Enable the AWS Config managed rule `restricted-ssh` and configure it to generate an Amazon SNS notification when a noncompliant rule is created or modified.
This is the most efficient solution as it uses a pre-built managed rule in AWS Config and integrates directly with SNS for notifications, minimizing operational overhead.
- ☐ Create an IAM role with broad permissions to manage security groups and network ACLs. Create an Amazon SNS topic to generate a notification whenever the role is assumed by a user.
This solution is not directly related to the specific requirement of monitoring SSH access from 0.0.0.0/0. Also, granting such broad permissions introduces significant security risks.
- ☐ Implement a service control policy (SCP) to prevent non-administrative users from creating or modifying security groups. Create a notification in the ticketing system when a user requests a rule that requires administrator permissions.

This approach restricts user access and introduces a manual approval process, which increases operational overhead and does not directly address the automated monitoring and notification requirement.

Overall Explanation:

The most efficient solution is to leverage the AWS Config managed rule `restricted-ssh`. This pre-built rule automatically checks for security groups that violate the specified policy and can trigger an Amazon SNS notification when a noncompliant rule is detected. This approach requires minimal configuration and maintenance, reducing operational overhead. Options A, C, and D involve either custom code development, unnecessary permission granting, or increased manual processes, making them less efficient.

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Question 56

Not Attempted

[Application Integration](#)

A company has deployed an application in an AWS account. The application consists of microservices that run on AWS Lambda and Amazon Elastic Kubernetes Service (Amazon EKS). A separate team supports each microservice. The company has multiple AWS accounts and wants to give each team its own account for its microservices. A solutions architect needs to design a solution that will provide service-to-service communication over HTTPS (port 443). The solution also must provide a service registry for service discovery. Which solution will meet these requirements with the LEAST administrative overhead?

- ☐ Create an inspection VPC. Deploy an AWS Network Firewall firewall to the inspection VPC. Attach the inspection VPC to a new transit gateway. Route VPC-to-VPC traffic to the inspection VPC. Apply firewall rules to allow only HTTPS communication.
This solution requires managing a firewall and routing rules, adding administrative overhead.
- ☒ Create a VPC Lattice service network. Associate the microservices with the service network. Define HTTPS listeners for each service. Register microservice compute resources as targets. Identify VPCs that need to communicate with the services. Associate those VPCs with the service network.
VPC Lattice simplifies service-to-service communication with built-in service discovery and HTTPS support, minimizing administrative overhead.

- ⊗ Create a Network Load Balancer (NLB) with an HTTPS listener and target groups for each microservice. Create an AWS PrivateLink endpoint service for each microservice. Create an interface VPC endpoint in each VPC that needs to consume that microservice. This solution involves managing multiple NLBs and PrivateLink endpoints, increasing administrative complexity.

- ⊗ Create peering connections between VPCs that contain microservices. Create a prefix list for each service that requires a connection to a client. Create route tables to route traffic to the appropriate VPC. Create security groups to allow only HTTPS communication.

This solution requires managing peering connections, route tables, and security groups across multiple VPCs, adding significant administrative overhead.

Overall Explanation:

VPC Lattice is designed to simplify service networking for service-to-service communication. It provides a managed service network with built-in service discovery and supports HTTPS, which directly addresses the requirements with the least amount of administrative overhead. The other options involve more manual configuration and management of network infrastructure.

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Question 57

Not Attempted

Databases



A company has a mobile game that reads most of its metadata from an Amazon RDS DB instance. As the game increased in popularity, developers noticed slowdowns related to the game's metadata load times. Performance metrics indicate that simply scaling the database will not help. A solutions architect must explore all options that include capabilities for snapshots, replication, and sub-millisecond response times. What should the solutions architect recommend to solve these issues?

- ⊗ Migrate the database to Amazon Aurora with Aurora Replicas.
While Aurora offers improved performance and read replicas, the problem statement indicates that scaling the database is not the solution. This option also requires significant effort.
- ⊗ Migrate the database to Amazon DynamoDB with global tables.

Migrating to DynamoDB could provide the necessary performance, but this would be a major application refactoring effort and might not be the most cost-effective or quickest solution. Although DynamoDB provides low latency, the question indicates that snapshots and replication are needed which are features of ElastiCache.

- ✔ Add an Amazon ElastiCache for Redis layer in front of the database.

ElastiCache for Redis offers sub-millisecond response times and supports replication and snapshots, making it a suitable caching layer to reduce the load on the RDS database. It can significantly improve metadata retrieval performance.

- ✘ Add an Amazon ElastiCache for Memcached layer in front of the database.

While ElastiCache for Memcached provides fast access, it lacks built-in persistence and snapshots, which are requirements outlined in the problem statement.

Overall Explanation:

Adding an ElastiCache for Redis layer in front of the database addresses the performance issues without requiring a database migration. Redis supports sub-millisecond response times and includes features for replication and snapshots, meeting all the requirements. Using Redis as a cache invalidates the need to scale the existing database.

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Question 58

Not Attempted

Security



A company uses AWS Organizations for its multi-account AWS setup. The security organizational unit (OU) of the company needs to share approved Amazon Machine Images (AMIs) with the development OU. The AMIs are created by using AWS Key Management Service (AWS KMS) encrypted snapshots. What are two solutions that will meet these requirements?

- ✔ Add the development team's OU Amazon Resource Name (ARN) to the launch permission list for the AMIs.

Adding the development OU's ARN to the launch permissions allows instances in that OU to be launched from the AMI.

- ✘ Add the Organizations root Amazon Resource Name (ARN) to the launch permission list for the AMIs.

Adding the root ARN is too broad and is not the best practice as it grants access to all accounts in the organization.

- ☒ Update the key policy to allow the development team's OU to use the AWS KMS keys that are used to decrypt the snapshots.

The KMS key policy must be updated to allow the development OU to use the key to decrypt the underlying snapshots when launching an EC2 instance from the AMI.

- ☐ Add the development team's account Amazon Resource Name (ARN) to the launch permission list for the AMIs.

While this works, it requires adding the ARN for each individual account, which is not ideal when dealing with an OU. Option A is preferred.

- ☐ Recreate the AWS KMS key. Add a key policy to allow the Organizations root Amazon Resource Name (ARN) to use the AWS KMS key.

Recreating the KMS key is not necessary. You can modify the existing key policy and granting access to the organization root is too broad.

Overall Explanation:

To share KMS-encrypted AMIs with another OU, two things are required. First, you need to allow the OU to use the KMS key to decrypt the snapshots which requires updating the KMS key policy. Second, you need to allow the OU to launch the AMI by adding the OU's ARN to the launch permission list for the AMIs. Therefore, options A and C are correct.

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Question 59

Not Attempted

Cost Optimization



A data analytics company has 80 offices that are distributed globally. Each office hosts 1 PB of data and has between 1 and 2 Gbps of internet bandwidth. The company needs to perform a one-time migration of a large amount of data from its offices to Amazon S3. The company must complete the migration within 4 weeks. Which solution will meet these requirements MOST cost-effectively?

- ☐ Establish a new 10 Gbps AWS Direct Connect connection to each office. Transfer the data to Amazon S3.

Establishing Direct Connect to each office would be very expensive and might not be feasible within the 4-week timeframe.

- ☒ Use multiple AWS Snowball Edge storage-optimized devices to store and transfer the data to Amazon S3.

Using multiple Snowball Edge devices is cost-effective and suitable for geographically distributed data. Snowball Edge is designed for transferring data when network bandwidth is limited.

- ☐ Use an AWS Snowmobile to store and transfer the data to Amazon S3.

Snowmobile is designed for very large datasets (10PB or more) in a single location, which is not the case here as the data is distributed across 80 offices. Furthermore, AWS has stopped offering Snowmobile as a service.

- ☐ Set up an AWS Storage Gateway Volume Gateway to transfer the data to Amazon S3.

Storage Gateway relies on internet bandwidth. With 1-2 Gbps bandwidth, transferring 1 PB from each of 80 offices in 4 weeks may not be feasible or cost-effective.

Overall Explanation:

Given the limited bandwidth and distributed nature of the data, using multiple AWS Snowball Edge devices is the most cost-effective solution. Direct Connect would be too expensive, Snowmobile is intended for large datasets in single location and has been discontinued, and Storage Gateway's performance is limited by the available internet bandwidth. Therefore option B is correct.

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Question 60

Not Attempted

Storage



A company has an Amazon Elastic File System (Amazon EFS) file system that contains a reference dataset. The company has applications on Amazon EC2 instances that need to read the dataset. However, the applications must not be able to change the dataset. The company wants to use IAM access control to prevent the applications from being able to modify or delete the dataset. Which solution will meet these requirements?

- ☐ Mount the EFS file system in read-only mode from within the EC2 instances.

Mounting in read-only mode doesn't utilize IAM access control.

- ☒ Create a resource policy for the EFS file system that denies the `elasticfilesystem:ClientWrite` action to the IAM roles that are attached to the EC2

instances.

This is the best solution. It uses IAM to specifically deny write access to the EFS file system for the EC2 instances' roles, preventing modifications or deletions.

- ☐ Create an identity policy for the EFS file system that denies the `elasticfilesystem:ClientWrite` action on the EFS file system.
Identity policies are attached to IAM users, groups, or roles, not directly to EFS file systems. Resource policies are used to control access to the EFS resource.
- ☐ Create an EFS access point for each application. Use Portable Operating System Interface (POSIX) file permissions to allow read-only access to files in the root directory.
POSIX file permissions at the root level may not be sufficient to prevent all modifications and don't leverage IAM for access control as requested.

Overall Explanation:

The best solution is to create a resource policy for the EFS file system that denies the `elasticfilesystem:ClientWrite` action to the IAM roles associated with the EC2 instances. This directly addresses the requirement of using IAM access control to prevent modifications and deletions. Options A and D don't utilize IAM and might not provide sufficient protection. Option C is incorrect because identity policies are not directly applicable to EFS.

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Question 61

Not Attempted

Databases



A company needs to retain user transaction data in an Amazon DynamoDB table for 7 years. Which solution is the MOST operationally efficient way to meet this requirement?

- ☐ Use DynamoDB point-in-time recovery (PITR) to continuously back up the table.
DynamoDB PITR provides recovery to any point in time within a 35-day window, which does not meet the 7-year retention requirement.
- ☒ Use AWS Backup to create backup schedules and retention policies for the table.
AWS Backup allows you to create backup schedules and retention policies for DynamoDB tables, ensuring data is retained for the required period (up to 100 years). It's a fully managed service that automates the backup process and reduces operational overhead.

- ⊗ Create an on-demand backup of the table by using the DynamoDB console. Store the backup in an Amazon S3 bucket. Set an S3 Lifecycle configuration for the S3 bucket.

While on-demand backups can be stored in S3, manually creating and managing these backups and the S3 lifecycle configuration is less operationally efficient than using AWS Backup.

- ⊗ Create an Amazon EventBridge (Amazon CloudWatch Events) rule to invoke an AWS Lambda function. Configure the Lambda function to back up the table and to store the backup in an Amazon S3 bucket. Set an S3 Lifecycle configuration for the S3 bucket.

This solution requires custom code and infrastructure, increasing operational overhead compared to using a managed backup service like AWS Backup. It also involves more management of individual components.

Overall Explanation:

AWS Backup provides the most operationally efficient solution because it is a fully managed service specifically designed for backing up and restoring AWS resources, including DynamoDB tables. It simplifies the process of creating backup schedules and retention policies to meet compliance requirements without requiring significant manual intervention. DynamoDB PITR only keeps backups for 35 days. The other options require more manual configuration and maintenance.

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Question 62

Not Attempted

Security



A company has hired an external vendor to perform work in the company's AWS account. The vendor uses an automated tool that is hosted in an AWS account that the vendor owns. The vendor does not have IAM access to the company's AWS account. The company needs to grant the vendor access to the company's AWS account in the MOST secure manner.

- ✔ Create an IAM role in the company's account to delegate access to the vendor's IAM role. Attach the appropriate IAM policies to the role for the permissions that the vendor requires.

This is the most secure option. IAM roles enable you to grant specific permissions to entities in other AWS accounts without sharing credentials.

- ⊗ Create an IAM user in the company's account with a password that meets the password complexity requirements. Attach the appropriate IAM policies to the user for the

permissions that the vendor requires.

Creating an IAM user for the vendor is less secure as it involves managing credentials directly within the company's account, which can be risky.

- ⊗ Create an IAM group in the company's account. Add the automated tool's IAM user from the vendor account to the group. Attach the appropriate IAM policies to the group for the permissions that the vendor requires.

You can't directly add an IAM user from another account to a group in your account. This approach is not feasible.

- ⊗ Create an IAM user in the company's account that has a permission boundary that allows the vendor's account. Attach the appropriate IAM policies to the user for the permissions that the vendor requires.

While permission boundaries add a layer of control, creating an IAM user is still less secure than using IAM roles for cross-account access.

Overall Explanation:

The most secure way to grant access to an external vendor's tool is to create an IAM role in the company's account. This role will be configured with the necessary permissions and allow the vendor's IAM role in their account to assume it. This approach avoids sharing credentials and follows the principle of least privilege. Options B, C and D are either insecure or not feasible.

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Question 63

Not Attempted

Cost Optimization



A company wants to run its experimental workloads in the AWS Cloud. The company has a set budget for cloud spending. The company's CFO is concerned about cloud spending accountability for each department and wants to receive a notification when the spending threshold reaches 60% of the budget. Which solution will meet these requirements?

- ✔ Use cost allocation tags on AWS resources to label owners. Create usage budgets in AWS Budgets. Add an alert threshold to receive notification when spending exceeds 60% of the budget.

This solution uses cost allocation tags for accountability and AWS Budgets with alert thresholds for notifications, meeting the requirements.

- ⊗ Use AWS Cost Explorer forecasts to determine resource owners. Use AWS Cost Anomaly Detection to create alert threshold notifications when spending exceeds 60% of the budget.

AWS Cost Explorer forecasts are not used to determine resource owners. Cost Anomaly Detection does not allow setting thresholds for specific percentages of a budget.

- ⊗ Use cost allocation tags on AWS resources to label owners. Use AWS Support API on AWS Trusted Advisor to create alert threshold notifications when spending exceeds 60% of the budget.

AWS Trusted Advisor via AWS Support API doesn't directly provide budget alert threshold notifications at a specific percentage.

- ⊗ Use AWS Cost Explorer forecasts to determine resource owners. Create usage budgets in AWS Budgets. Add an alert threshold to receive notification when spending exceeds 60% of the budget.

AWS Cost Explorer forecasts are not used to determine resource owners. AWS Budgets can handle budget and threshold notifications, but not using cost explorer forecasts.

Overall Explanation:

The correct solution is to use cost allocation tags for accountability and AWS Budgets with alert thresholds for notifications at 60% of the budget. AWS Budgets allows you to set up email notifications based on the actual or forecasted cost or usage exceeding your defined budget thresholds. Cost allocation tags can be used to track spend at a granular level such as department.

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Question 64

Not Attempted

Networking



A company wants to deploy an internal web application on AWS. The web application must be accessible only from the company's office. The company also needs to download security patches for the web application from the internet. The company has established a VPC and a Site-to-Site VPN connection to their office. A solutions architect needs to design a secure architecture to meet these requirements. Which solution is the MOST appropriate?

- ⊗ Deploy the web application on Amazon EC2 instances in public subnets behind a public Application Load Balancer (ALB). Attach an internet gateway to the VPC. Configure the ALB's security group to allow inbound traffic from 0.0.0.0/0.

This solution exposes the application to the public internet, which is not a secure architecture and does not meet the requirement of only being accessible from the company's office.

- ✔ Deploy the web application on Amazon EC2 instances in private subnets behind an internal Application Load Balancer (ALB). Deploy NAT gateways in public subnets. Attach an internet gateway to the VPC. Configure the ALB's security group to allow inbound traffic from the company's office network CIDR block.

This is the most suitable solution. The internal ALB ensures the application is not publicly accessible. The EC2 instances residing in the private subnet are not directly exposed to the internet, increasing security. The NAT Gateway allows the EC2 instances to download patches from the internet. Security group rules are properly configured.

- ⊗ Deploy the web application on Amazon EC2 instances in public subnets behind an internal Application Load Balancer (ALB). Deploy NAT gateways in private subnets. Attach an internet gateway to the VPC. Configure the ALB's security group to allow outbound traffic to the company's office network CIDR block.

Placing the EC2 instances in public subnets negates the benefit of the VPN connection for security. NAT Gateways in private subnets does not allow the EC2 instances to download patches. Also, the outbound rule on the ALB's security group is incorrect.

- ⊗ Deploy the web application on Amazon EC2 instances in private subnets behind a public Application Load Balancer (ALB). Attach an internet gateway to the VPC. Configure the ALB's security group to allow outbound traffic to 0.0.0.0/0.

A public ALB would expose the web application to the internet, conflicting with the requirement to limit access to the company's office. In addition, outbound rules from the ALB do not control access to the application.

Overall Explanation:

The correct solution is to deploy the EC2 instances in private subnets behind an internal Application Load Balancer (ALB). This ensures that the application is only accessible from within the VPC and through the VPN connection to the company's office. The NAT Gateway in the public subnet allows the instances to access the internet for downloading necessary patches, while maintaining their private nature. The ALB security group inbound rule allowing the CIDR block of the company's office network makes sure that only connections from that network are accepted by the ALB.

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Question 65

Not Attempted

Databases



A company maintains its accounting records in a custom application that runs on Amazon EC2 instances. The company needs to migrate the data to an AWS managed service for development and maintenance of the application data. The solution must require minimal operational support and provide immutable, cryptographically verifiable logs of data changes. Which solution will meet these requirements MOST cost-effectively?

- ☐ Copy the records from the application into an Amazon Redshift cluster.
Amazon Redshift is a data warehouse service and is not designed for immutable, cryptographically verifiable logs of data changes.
- ☐ Copy the records from the application into an Amazon Neptune cluster.
Amazon Neptune is a graph database service and is not designed for immutable, cryptographically verifiable logs of data changes.
- ☐ Copy the records from the application into an Amazon Timestream database.
Amazon Timestream is a time-series database service and is not designed for immutable, cryptographically verifiable logs of data changes.
- ☒ Copy the records from the application into an Amazon Quantum Ledger Database (Amazon QLDB) ledger.
Amazon QLDB is a fully managed ledger database that provides a transparent, immutable, and cryptographically verifiable transaction log owned by a central trusted authority. It requires minimal operational support.

Overall Explanation:

Amazon Quantum Ledger Database (Amazon QLDB) is a fully managed ledger database that provides a transparent, immutable, and cryptographically verifiable transaction log. This meets the requirements for minimal operational support and immutable logs of data changes. Therefore, it's the most cost-effective solution. QLDB is designed to review the history of all changes made to the application data over time.

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